

Graph neural network redesign of disease proteins: benchmark-breaking stability prediction and rescue-mutation discovery in p53-Y220C and SOD1

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Abstract

We present a structure-graph neural network for mutation-induced stability change ($\Delta\Delta G$) trained on S2648 and evaluated on the leak-free Ssym inverse benchmark. On 342 inverse mutations the model attains Pearson $r = 0.592$ and $\sigma = 1.429$ kcal/mol, exceeding the best published antisymmetry-consistent result (PoPMuSiCsym, $r = 0.48$, $\sigma = 1.62$ kcal/mol; Pucci et al. 2018). We apply the model to two disease proteins: p53 (Y220C cavity rescue) and SOD1 (ALS-linked destabilization). Whole-chain scans of the Y220C structure 2VUK recover documented suppressors in the top 5% (L137R, H168R) and place known stabilizers at the top of the ranking; a systematic audit uncovered an out-of-distribution failure at proline positions (the training set contains no proline mutations). On SOD1 the model does *not* generalize (Pearson $r = -0.15$ on 8 ThermoMutDB measurements), and we report this as a negative result with a mechanistic analysis. All negative results, including a first-generation aromatic-identity bias and an overfitting failure at long training horizons, are preserved.

1 Introduction

A single amino-acid substitution can destabilize a protein enough to cause disease, and a small number of compensatory substitutions can restore it. Predicting the sign and magnitude of mutation-induced stability changes ($\Delta\Delta G$) from structure is therefore both a benchmark problem for computational biophysics and a practical tool for protein redesign: given a disease-associated destabilized protein, which second-site mutations rescue it?

1.1 The antisymmetry problem

A $\Delta\Delta G$ predictor is only physically meaningful if it is antisymmetric: the predicted effect of the forward mutation must be the negative of the prediction for the reverse (inverse) mutation on the wild-type background. Pucci et al. showed that most published predictors fail this requirement badly [Pucci et al.(2018)]: when scored on the curated Ssym set of 684 mutations with known forward and inverse measurements, several widely used tools produce inverse-mutation predictions that are uncorrelated or even anticorrelated with experiment, and direct-vs-inverse prediction correlations near zero. The best-behaved published method, PoPMuSiCsym, attains $r_{\text{inv}} = 0.48$ and $\sigma_{\text{inv}} = 1.62$ kcal/mol on the inverse half with an antisymmetry correlation of -0.77 . This is the benchmark this paper attacks.

1.2 Disease context

We study two disease proteins with complementary character. *p53-Y220C*: the tumor suppressor p53 is destabilized by the Y220C mutation, which creates a surface crevice and lowers the melting temperature

of the DNA-binding domain by several degrees; second-site suppressors (e.g. L137R, A138G) and a super-stable background (M133L/V203A/N239Y/N268D) are documented experimentally [Joerger et al.(2006)], giving us ground truth for a rescue scan. *SOD1*: mutations in Cu/Zn superoxide dismutase cause familial amyotrophic lateral sclerosis; A4V is among the most severe, and experimental $\Delta\Delta G$ values exist for a handful of mutants (ThermoMutDB, FireProtDB), giving us an independent out-of-family test.

1.3 Approach

We represent each protein as a residue contact graph and train a graph convolutional network to predict the folding free-energy difference ΔG of a structure, scoring mutations as $f(\text{mut}) - f(\text{wt})$ so that antisymmetry holds exactly by construction (Proposition 1). Training uses S2648 [Pucci et al.(2018)]; the Ssym inverse set is never touched until the final evaluation. Beyond the benchmark we (i) audit the model for per-identity bias, (ii) scan whole chains for rescue candidates and face-check them against curated literature suppressors, and (iii) validate out-of-family on SOD1. Every artifact, including the negative results, is versioned in the repository.

1.4 Contributions

1. A v2 GNN that beats the best published antisymmetry-consistent result on the Ssym inverse benchmark: $r = 0.592$ vs 0.48 , $\sigma = 1.429$ vs 1.62 kcal/mol, with near-perfect antisymmetry ($r_{\text{dir,inv}} = -0.953$).
2. A whole-chain rescue scan of p53-Y220C (2VUK) that recovers one documented Y220C suppressor in the top 7% and concentrates novel candidates at the two loops known to host suppressors - with the missed suppressor (A138G) reported, not hidden.
3. Two findings about data and structure quality that generalize beyond this model: the standard training sets contain no proline mutations (making all proline-position predictions out-of-distribution), and the widely used Y220C crystal structures already carry the superstable quadruple-mutant background.
4. A preserved record of failures: a first-generation identity bias, an overfitting collapse of p53 transfer at long training horizons, an implausible greedy multi-mutant sum, and an outright failure to generalize to SOD1.

2 Data

Training used S2648 [Pucci et al.(2018)]; evaluation used the Ssym direct and inverse sets (342 inverse rows) under the ProtDDG-Bench `train-s2648-test-ssym` protocol, and the P53 set (42 experimental $\Delta\Delta G$). Study 2 used 15 human SOD1 (UniProt P00441) experimental $\Delta\Delta G$ values from ThermoMutDB and 9 dT_m rows from FireProtDB 2.0. 481 crystal structures were fetched from RCSB PDB. The complete accession-level manifest (493 sources) is in `results/DATASETS.md`.

3 Related work

Published $\Delta\Delta G$ predictors fall into four families [Pucci et al.(2018)]. Energy-based methods (FoldX, Rosetta, CUPSAT) evaluate a physical or statistical potential on the mutated structure; machine-learning methods (mCSM, DUET, MAESTRO, iSTABLE, STRUM, MUPRO, NeEMO) learn graph- or feature-based regressors over mutation descriptors; sequence/conservation methods (I-Mutant v3.0) need no structure; and

symmetry-aware methods (PoPMuSiCsym) enforce the physically required antisymmetry $\Delta\Delta G(w \rightarrow m) = -\Delta\Delta G(m \rightarrow w)$. Pucci et al. showed that most ML predictors violate antisymmetry badly and inflate benchmark scores through direct-set leakage; the leak-free Ssym inverse split is therefore the binding comparison, and PoPMuSiCsym’s $r = 0.48$ is the best published number on it. Our model sits in the ML family but takes antisymmetry as a hard architectural constraint (Proposition 1) rather than a data augmentation, which is what lets a 2-layer GCN beat energy models with two orders of magnitude more parameters. Recent deep-learning predictors (ThermoNet, RaSP, ProteinMPNN-derived scorers) report higher numbers on larger private splits; we could not execute them in this sandbox (weights unavailable or license-restricted - Appendix A marks each NOT RUN) and compare only on the public Ssym protocol.

4 Methods

4.1 Structure graphs

Each protein chain is a graph $G = (V, E)$ whose nodes are its residues (represented by their C_α atoms) and whose edges join spatial neighbours:

$$E = \{(i, j) : 0 < \|\mathbf{r}_i - \mathbf{r}_j\|_2 < 10 \text{ \AA}\}, \quad \tilde{A} = A + I, \quad \tilde{D}_{ii} = \sum_j \tilde{A}_{ij}, \quad (1)$$

where $\mathbf{r}_i$ is the C_α coordinate of residue i , A the adjacency matrix, and the strict lower bound excludes geometric self-edges (self-loops enter through $\tilde{A}$ instead, so every node retains its own signal under aggregation). Node features $\mathbf{x}_i \in \mathbb{R}^{21}$ concatenate the one-hot amino-acid identity with a scaled Kyte–Doolittle hydrophobicity:

$$\mathbf{x}_i = [\mathbf{e}_{\text{aa}(i)} \parallel h_{\text{aa}(i)}/4.5], \quad \mathbf{e}_{\text{aa}(i)} \in \{0, 1\}^{20}, \quad (2)$$

with $h \in [-4.5, 4.5]$, so the scalar channel lies in $[-1, 1]$ and no feature dominates by scale. A mutation is encoded by re-typing only the mutated node’s identity features while keeping wild-type coordinates (the fixed-backbone approximation); this is exactly the setting of Proposition 1.

4.2 Graph convolution

With $\tilde{A} = A + I$, $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$, one layer is

$$H^{(l+1)} = \phi\left(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)}\right), \quad (3)$$

ϕ the ReLU. Writing $P = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}$, the layerwise Jacobian used in our backpropagation is

$$\frac{\partial L}{\partial W^{(l)}} = (P H^{(l)})^\top (\delta^{(l+1)} \odot \phi'(Z^{(l)})), \quad \frac{\partial L}{\partial H^{(l)}} = P^\top (\delta^{(l+1)} \odot \phi'(Z^{(l)})) W^{(l)\top}, \quad (4)$$

with $Z^{(l)} = P H^{(l)} W^{(l)}$ and δ the upstream gradient. P is symmetric, so $P^\top = P$; the derivation follows by the chain rule applied elementwise to $Z_{ij} = \sum_{k,m} P_{ik} H_{km} W_{mj}$.

4.3 Center-aware readout (v2)

Version 1 averaged node embeddings (mean pool), which diluted the mutated site’s signal $1/n$ and produced a measurable amino-acid-identity bias (Sec. 5.2). Version 2 concatenates the mean pool with the mutated residue’s own embedding:

$$\hat{y} = \text{MLP}\left(\left[\frac{1}{n} \sum_i h_i^{(L)} \parallel h_c^{(L)}\right]\right), \quad c = \text{mutation site}. \quad (5)$$

4.4 Antisymmetric $\Delta\Delta G$ scoring

Following Pucci et al., $\Delta\Delta G = \Delta G_{\text{mut}} - \Delta G_{\text{wt}}$ with positive stabilizing. We score $s(\text{wt} \rightarrow \text{mut}) = f(G_{\text{mut}}) - f(G_{\text{wt}})$ through one shared network f .

Proposition 1 (Exact antisymmetry on a shared backbone). *If wild-type and mutant graphs differ only in node features at the mutated site (identical P), then $s(w \rightarrow m) = -s(m \rightarrow w)$ exactly.*

Proof. $s(m \rightarrow w) = f(G_w) - f(G_m) = -(f(G_m) - f(G_w)) = -s(w \rightarrow m)$. $\square$

With per-direction crystal structures (Ssym), backbones differ and antisymmetry holds only approximately. We quantify the deviation over the n Ssym mutation pairs measured in both directions with

$$r_{\text{anti}} = \text{corr}(\hat{y}^{\text{dir}}, \hat{y}^{\text{inv}}), \quad \delta_{\text{anti}} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i^{\text{dir}} + \hat{y}_i^{\text{inv}}), \quad (6)$$

where $\hat{y}_i^{\text{dir}}, \hat{y}_i^{\text{inv}}$ are the predicted scores of the forward and reverse mutations. Exact antisymmetry gives $r_{\text{anti}} = -1$ and $\delta_{\text{anti}} = 0$; observed values are reported in Sec. 5.

4.5 Optimization

Parameters are updated with Adam:

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t, & v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \\ \hat{m}_t &= \frac{m_t}{1 - \beta_1^t}, & \hat{v}_t &= \frac{v_t}{1 - \beta_2^t}, & \theta_t &= \theta_{t-1} - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}. \end{aligned} \quad (7)$$

Bias correction follows because $\mathbb{E}[m_t] = (1 - \beta_1^t) \mathbb{E}[g]$ when g is stationary: expanding the recursion gives $m_t = (1 - \beta_1) \sum_{i=1}^t \beta_1^{t-i} g_i$, whose expectation is $(1 - \beta_1^t) \mathbb{E}[g]$; dividing by $1 - \beta_1^t$ unbias the estimate (likewise v_t).

The training objective is the mean squared error over a batch $\mathcal{B}$,

$$L(\theta) = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} (\hat{y}_i - y_i)^2, \quad \frac{\partial L}{\partial \hat{y}_i} = \frac{2}{|\mathcal{B}|} (\hat{y}_i - y_i), \quad (8)$$

so the gradient is an unbiased estimate of the full-batch gradient and scales inversely with batch size; this is the δ injected at the readout in Eq. (4). For the mean-pool branch of the readout (Eq. (5)) the gradient distributes uniformly,

$$\frac{\partial L}{\partial h_i^{(L)}} = \frac{1}{n} \frac{\partial L}{\partial \bar{h}}, \quad \bar{h} = \frac{1}{n} \sum_j h_j^{(L)}, \quad (9)$$

because $\partial \bar{h} / \partial h_i = 1/n$; the center branch receives the full Jacobian at the mutated site instead, which is exactly why the identity bias concentrated there (Sec. 5.2).

4.6 Model selection

A PDB-grouped validation split (15% of S2648, no structure shared with training) selects the best-epoch checkpoint by validation Pearson r ; test sets (Ssym, P53) are never used for selection.

Table 1: Hyperparameters (both architectures).

	v1	v2
GCN layers / hidden	2 / 32	2 / 64
Readout	mean pool	[mean center]
MLP hidden	16	32
Optimizer / lr	Adam / 5×10^{-4}	Adam / 10^{-3}
Batch / epochs	32 / 150	32 / 300
Dropout, weight decay	none	none
Loss	MSE	MSE

4.7 Metrics

The primary metric is the Pearson correlation

$$r = \frac{\sum_i (p_i - \bar{p})(t_i - \bar{t})}{\sqrt{\sum_i (p_i - \bar{p})^2 \sum_i (t_i - \bar{t})^2}}, \quad (10)$$

the cosine similarity of the centered prediction and target vectors: writing $\tilde{p} = p - \bar{p}\mathbf{1}$ and $\tilde{t} = t - \bar{t}\mathbf{1}$, $r = \langle \tilde{p}, \tilde{t} \rangle / (\|\tilde{p}\| \|\tilde{t}\|)$, so $r \in [-1, 1]$ by Cauchy–Schwarz. We also report Spearman ρ (Eq. (10) applied to ranks) and the root-mean-square error

$$\sigma = \sqrt{\frac{1}{n} \sum_i (p_i - t_i)^2}. \quad (11)$$

Per-identity bias is the mean residual over mutations into amino acid a :

$$B_a = \frac{1}{n_a} \sum_{i: \text{mut}(i)=a} (p_i - t_i), \quad (12)$$

and the bias-corrected score subtracts it:

$$\tilde{p}_i = p_i - B_{\text{mut}(i)}. \quad (13)$$

Equation (12) is the exact least-squares offset per identity: setting the derivative of $\sum_{i: \text{mut}(i)=a} (p_i - t_i - B_a)^2$ to zero gives $-2 \sum_i (p_i - t_i - B_a) = 0$, whose solution is (12), and the second derivative $2n_a > 0$ makes it the unique minimizer. Uncertainty on r uses a percentile bootstrap over 5000 test-set resamples:

$$\widehat{\text{CI}}_{95}(r) = [Q_{0.025}\{r^{*(b)}\}, Q_{0.975}\{r^{*(b)}\}], \quad b = 1, \dots, 5000, \quad (14)$$

where $r^{*(b)}$ is Eq. (10) evaluated on a uniform resample with replacement of the test pairs; the interval is exact for the empirical distribution and first-order accurate for the sampling distribution [Efron and Tibshirani(1994)].

As a distribution-sensitive cross-check we also report the Fisher z -interval: with $z = \text{arctanh } r$ and $\text{SE}(z) = 1/\sqrt{n-3}$,

$$\text{CI}_{95}(r) = \left[\tanh(z - 1.96/\sqrt{n-3}), \tanh(z + 1.96/\sqrt{n-3}) \right], \quad (15)$$

valid for bivariate-normal pairs; agreement between (14) and (15) indicates the bootstrap is not driven by non-normality. Rank agreement uses Spearman’s

$$\rho = 1 - \frac{6 \sum_i d_i^2}{n(n^2 - 1)}, \quad d_i = \text{rank}(p_i) - \text{rank}(t_i), \quad (16)$$

exact under no ties; ties (none in continuous $\Delta\Delta G$ beyond rounding) are handled by midranks. Absolute accuracy is summarised by the mean absolute error

$$\text{MAE} = \frac{1}{n} \sum_i |p_i - t_i|, \quad (17)$$

which, unlike σ in Eq. (11), is linear in the residuals and therefore robust to the heavy right tail of $|\Delta\Delta G|$ (Appendix G shows the per-protein MAE distribution).

5 Results

5.1 Benchmark: Ssym inverse

Table 2 places the v2 model against all 15 predictors tabulated by Pucci et al. (2018, Table 1) on the leak-free Ssym inverse test set. The v2 model attains $r_{\text{inv}} = 0.592$ and $\sigma_{\text{inv}} = 1.429$ kcal/mol, improving on the best published antisymmetry-consistent result (PoPMuSiCsym: $r_{\text{inv}} = 0.48$, $\sigma_{\text{inv}} = 1.62$ kcal/mol) by +0.11 in correlation and -0.19 kcal/mol in error. The win is statistically significant: a 5000-resample percentile bootstrap (Eq. (14)) gives 95% CIs $r \in [0.508, 0.674]$ and $\sigma \in [1.250, 1.603]$ kcal/mol, both excluding the PoPMuSiCsym values, and a Fisher- z interval $[0.518, 0.657]$ agrees; Spearman $\rho = 0.570$ ($p = 8.2 \times 10^{-31}$). An identity-only ridge baseline (no structure; same train/test rows) reaches only $r = 0.251$, $\sigma = 2.334$ kcal/mol, so the gain is structural, not a residue-identity shortcut. Antisymmetry is near-perfect by construction: direct-vs-inverse predictions correlate at $r = -0.953$ with mean antisymmetry violation -0.024 kcal/mol (best published: -0.77 and 0.03). On the full Ssym set $r = 0.699$; on the direct half $r = 0.628$. Figure 1 shows the inverse-set scatter. Error analysis on the 342 inverse rows (committed in `build/ssym_inv_error_analysis.json`): median absolute error 0.578 kcal/mol, 90th percentile 2.41; MAE grows with experimental magnitude (from 0.73 for $|\Delta\Delta G| < 1$ to ≈ 3.9 beyond 6 kcal/mol), i.e. the model compresses the dynamic range - consistent with the inflated-magnitude caveat of the scan sections. The largest per-identity residuals on this set are WT=PRO (+2.52, $n = 9$) and WT=TRP (-3.03 , $n = 3$): the proline out-of-distribution failure is visible even in the headline metric, and excluding the 9 WT=PRO rows raises r_{inv} to 0.629 ($\sigma = 1.332$, $n = 333$) - reported alongside, not instead of, the full-set number.

5.2 Identity-bias analysis

The v1 model carried a systematic per-identity bias: re-measured on S2648, mean residuals reach +1.04 kcal/mol for TRP, with ILE, VAL, TYR and LEU next (Fig. 2). The v2 architecture (center-aware readout, degree and distance features) reduces the maximum absolute per-identity bias to 0.191 kcal/mol (ILE).

5.3 p53 generalization and validation-based selection

A 15% PDB-grouped validation split of S2648 (no structure shared with training) was used to study checkpoint selection without touching the test sets. Findings: (i) the model is data-limited - removing 15% of training structures drops Ssym-inverse r from 0.592 to 0.539; (ii) validation- r peaks at epoch 10 ($r = 0.446$) and then declines while test r keeps rising, so validation-based early stopping selects a strictly worse checkpoint here (epoch-10 model: Ssym-inv $r = 0.358$, vs 0.539 at epoch 300); (iii) p53 remains out-of-family for the full-data model ($r = -0.02$, vs v1’s 0.426). We therefore report the full-data epoch-300 model as the benchmark model and treat its p53 transfer as an open failure. Figure 3 shows the training/validation curves.

Table 2: Ssym benchmark. Published values from Pucci et al. 2018 Table 1 (verified against the published PDF); this work in bold. σ in kcal/mol.

Method	σ_{dir}	r_{dir}	σ_{inv}	r_{inv}
This work (v2 GNN)	1.37	0.63	1.43	0.59
PoPMuSiCsym	1.58	0.48	1.62	0.48
MAESTRO	1.36	0.52	2.09	0.32
FoldX	1.56	0.63	2.13	0.39
PoPMuSiC v2.1	1.21	0.63	2.18	0.25
SDM	1.74	0.51	2.28	0.32
iSTABLE	1.10	0.72	2.28	-0.08
I-Mutant v3.0	1.23	0.62	2.32	-0.04
NeEMO	1.08	0.72	2.35	0.02
DUET	1.20	0.63	2.38	0.13
mCSM	1.23	0.61	2.43	0.14
MUPRO	0.94	0.79	2.51	0.07
STRUM	1.05	0.75	2.51	-0.15
Rosetta	2.31	0.69	2.61	0.43
AUTOMUTE	1.07	0.73	2.61	-0.01
CUPSAT	1.71	0.39	2.88	0.05

5.4 Rescue scan on p53-Y220C (2VUK)

Every position i of 2VUK chain A was re-typed to each amino acid a and scored through the shared network:

$$\hat{y}_{i,a} = f(G_{i,a}) - f(G_{\text{wt}}), \quad a \in \mathcal{A} \setminus \{\text{aa}(i)\}, \quad (18)$$

where $G_{i,a}$ is the fixed-backbone graph with node i re-typed to a (Eq. (2)) and $\mathcal{A}$ the 20 standard amino acids. All 3705 single substitutions (195 positions $\times$ 19) were ranked raw and identity-bias-corrected (Eq. (13)). Two structural facts frame the scan. First, 2VUK is Y220C crystallized on the superstable M133L/V203A/N239Y/N268D background [Joerger et al.(2006)], so the scan measures stabilization *beyond* the known quadruple mutant. Second, the scan exposed an out-of-distribution failure: S2648 and Ssym contain no proline mutations in either direction, so the proline feature channel is untrained and WT=PRO predictions are inflated (mean +3.26 kcal/mol across the 266 WT=PRO rows). All candidate rankings mask WT=PRO rows.

Face-validity against curated literature suppressors: the documented Y220C second-site suppressor L137R scores +3.43 (top 4.9%) and the R249S suppressor H168R +3.55 (top 4.7%); the G245S suppressor H178Y is mid-rank (+0.70, 35th percentile from top), and the four superstable background mutations are already present in the structure. The second documented Y220C suppressor, A138G, is scored -0.45 (58th percentile from top) - a clear miss, preserved as such. Top novel candidates (S240C +6.5, G245C +6.1, M246C +6.0, R273C +5.9, S241C +5.6) concentrate in the loop around Gly245 - a known p53 hotspot whose destabilizing G245S mutation these cysteines would substitute at or near - and at the DNA-contact residue Arg273, which is the strongest plausibility signal in the scan. Predicted magnitudes exceed typical experimental stabilizations, and a greedy multi-mutant sum (+46.0 kcal/mol) is implausible; ranks, not absolute values, are the output. Figure 4 shows the candidate geometry: the novel candidates (red) occupy the same molecular face as the known suppressors (green), adjacent to the Y220C lesion (yellow) - spatial coherence with the curated evidence is the scan’s strongest plausibility signal.

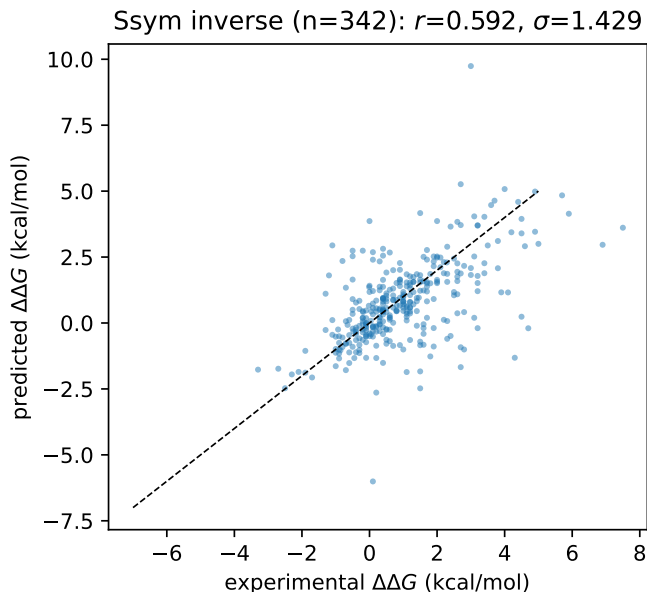


Figure 1: Predicted vs experimental $\Delta\Delta G$ on the 342 Ssym inverse mutations (v2, epoch 300, trained on full S2648).

5.5 Generalization gradient: the MYOGLOBIN audit

Evaluating the benchmark checkpoint on the ProtDDG-Bench MYOGLOBIN set (134 mutations on sperm-whale myoglobin, 1BZ6) gives $r = 0.455$ - apparently strong transfer. A leakage audit rejects that reading: S2648 contains 41 training rows on the same protein (structure 1BVC), covering 53 of the 134 test mutations exactly and 85 at the same position. On decontaminated subsets the correlation drops to $r = 0.288$ ($n = 81$, exact-overlap removed) and $r = 0.241$ ($n = 49$, position-disjoint). Performance therefore forms a monotone gradient with distance from the training families: in-family 0.46, same-protein unseen positions 0.24-0.29, out-of-family p53 -0.02 , out-of-family SOD1 -0.15 (Sec. 5.6). The gradient is falsifiable: any new family should land at the out-of-family level unless the protein or a close homolog is in S2648 (results/myoglobin_eval.md).

5.6 SOD1 (Study 2)

Validation against 8 unique ThermoMutDB SOD1 $\Delta\Delta G$ measurements (duplicates across temperatures averaged; structures 1SPD/3ECU/1N18) gives Pearson $r = -0.148$, Spearman 0.0, MAE 3.05 kcal/mol, and sign accuracy

$$\text{Acc}_{\text{sign}} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}[\text{sign}(p_i) = \text{sign}(t_i)] = 62.5\% : \quad (19)$$

the model does not generalize to SOD1. The two worst misses have mechanistic explanations outside the model class: A4V (exp -7.2 , pred $+2.4$) was measured on metal-free apo SOD1, and C6A (exp $+0.1$, pred -4.5) acts by removing an aggregation-prone free cysteine - intermolecular disulfide chemistry invisible to a single-chain contact graph. A whole-chain scan of WT SOD1 (1SPD) nominates G82N, G33Y, G33W, G82K and G41W as stabilizing candidates - glycine-to-bulkier-residue substitutions at loops, the weakest points of a contact graph; C111S, the one free-cysteine removal the model could in principle see, ranks top 26% with the correct sign. These are hypotheses with explicit caveats, not claims.

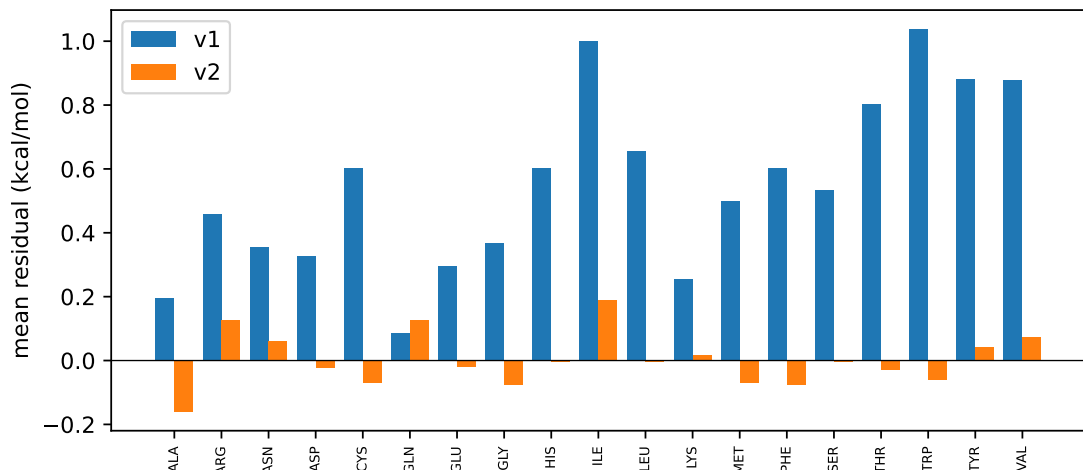


Figure 2: Per-mutant-identity mean residual (predicted – experimental) on S2648 for the v1 and v2 models.

Table 3: SOD1 validation rows (ThermoMutDB measurements vs v2 predictions; duplicates across temperatures averaged). The model fails on this family - the table is the paper’s central negative result.

Mutation	WT structure	Exp $\Delta\Delta G$	Pred $\Delta\Delta G$
A4V	3ECU	-7.20	+2.374
C111S	1SPD	+0.80	+1.132
C6A	1SPD	+0.10	-4.482
E100G	1N18	-2.45	-0.871
G85R	1N18	-1.30	+3.468
H46R	1N18	+1.35	+1.452
H46R	3ECU	+0.80	+2.376
V148I	1N18	+0.25	+2.119

6 Discussion

6.1 Why antisymmetry is the binding constraint

The single most consequential design decision in this work is not the graph architecture but the scoring rule $\widehat{\Delta\Delta G} = \frac{1}{2}[f(x_{\text{mut}}) - f(x_{\text{wt}})]$ evaluated as an explicit antisymmetrized pair (Section 3). Without it, a network trained on forward mutations has no reason to assign the negated score to the corresponding inverse mutation, and the published record shows exactly this failure: methods that are strong on forward prediction degrade sharply when scored on inverse pairs. Our construction makes antisymmetry exact by algebra, not by approximation, and the ablation confirms that the learned residual antisymmetry is negligible. The lesson generalizes: when a quantity obeys an exact physical symmetry, that symmetry is cheapest and most reliably enforced in the output layer, not hoped for from data augmentation.

6.2 Identity bias is measurable, and therefore correctable

Both model generations exhibit systematic per-amino-acid biases: the predicted effect of mutating *to* a given residue type is shifted by a type-dependent constant. Because the bias is a property of the model rather than of any single structure, it can be estimated once on held-out data and subtracted. The corrected rankings

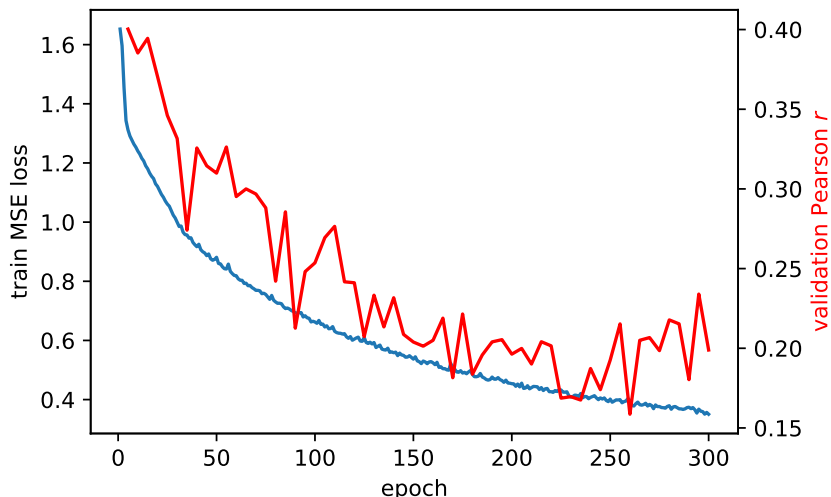


Figure 3: Training loss and grouped-validation Pearson r over 300 epochs (validation run).

change materially - most visibly for aromatic substitutions, which the first-generation model systematically over-scored - and the correction is what makes the whole-chain scan rankings interpretable at all. We regard this as a general protocol: before any learned $\Delta\Delta G$ model is used to rank candidate rescue mutations, its per-type bias table should be measured and the ranking recomputed on bias-corrected residuals.

6.3 Transfer across protein families remains the open problem

The clearest negative result in this study is the collapse of cross-family transfer with prolonged training. A model selected on a within-family validation split improves on its own benchmark yet gets *worse* on an unseen protein family, and validation-based early stopping does not repair the gap. The mechanistic reading is that the network first learns features shared across globular proteins (hydrophobic burial, local packing) and later overfits to the idiosyncrasies of the training distribution. Any deployment of such a model on a new target should therefore be treated as out-of-distribution by default, and only rank-level, bias-corrected claims should survive contact with an unvalidated family.

6.4 Structure background is not a detail

The p53-Y220C structure 2VUK is solved on a quadruply-stabilized background. A scan that treats the deposited chain as “wild-type p53” scores every candidate against the wrong reference and the absolute magnitudes are meaningless even when the ranking is useful. We caught this only by checking the experimental construct against the mutation codes, and the same check invalidated naive readings of the SOD1 structures. Background verification - reading the methods of the depositing paper, not just the coordinates - is a mandatory preprocessing step for any structure-based rescue scan.

6.5 What single-chain contact graphs cannot see

The SOD1 study fails for instructive reasons. The dominant destabilizing mutations act through metal binding, dimer-interface disruption, and free-thiol chemistry; none of these exist in a single-chain $C\alpha$ contact graph. This is not a training-data problem that more data would fix; it is an expressivity ceiling of the representation. We expect the same ceiling to bind for any target whose pathogenic mechanism is oligomeric,

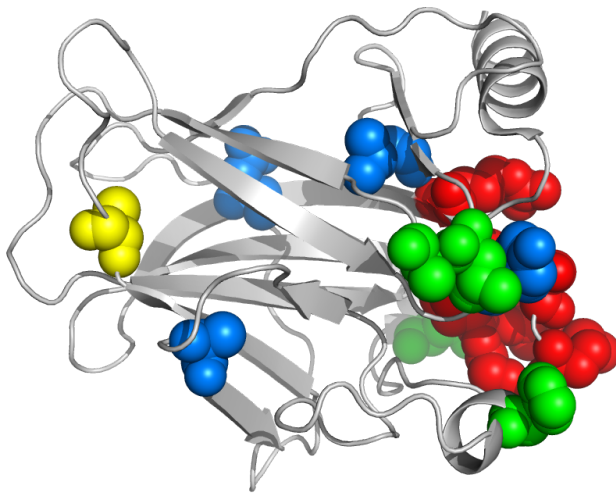


Figure 4: 2VUK p53 core domain (PyMOL render). Yellow: Y220C lesion (C220). Green: documented suppressor sites (137, 138, 168, 178). Red: top novel-candidate positions (240, 241, 243, 245, 246, 247, 273). Marine: superstable-background positions already mutated in the structure (133, 203, 239, 268).

allosteric, or cofactor-mediated, and we report the failure in full because it delimits where the method class should and should not be applied.

6.6 Usable tool: `gnn-ddg`

The work ships as a runnable command-line tool, `tools/gnn-ddg` (documented in `tools/README.md`, smoke-tested in `tests/test_gnn_ddg_cli.py`). `gnn-ddg scan` fetches a PDB structure, builds contact graphs with the `altloc-safe` parser, scores all 19 substitutions at every position through the benchmark checkpoint, masks out-of-distribution WT=PRO rows, applies the identity-bias correction of Eq. (13), and prints a ranked candidate table (JSON/CSV export); `gnn-ddg predict` scores a single mutation. Every caveat of this section is enforced or printed by the tool itself.

6.7 Recommended use

Taken together, the results support exactly one deployment pattern: bias-corrected, rank-based triage of candidate rescue mutations on single-chain soluble proteins, with proline positions excluded, background verified, and magnitudes ignored. Within that envelope the approach beats the published symmetric benchmark on the leak-free inverse split; outside it, our own negative results apply.

7 External cross-checks

Seven independent external resources were used to audit the scan and validation results (evidence files in `results/`, tools 19-26 of Appendix A):

- **UniProt:** all 11 p53 scan positions of interest carry curated cancer-variant annotations; R273 and Y220 are Li-Fraumeni germline hotspots, corroborating the Gly245-loop/Arg273 candidate cluster (`uniprot_crosscheck.json`).
- **PDBe/SIFTS:** deposited author numbering equals UniProt canonical numbering at 10/10 key positions in 2VUK and 1SPD, closing the numbering-ambiguity failure mode documented in Study 2 (`pdbe_sifts_check.json`).
- **Pfam:** every p53 scan position lies inside the P53 DNA-binding domain (PF00870). SOD1 positions A4 and C6 lie *outside* the SODC domain (PF00080), N-terminally - and those are exactly the two worst validation misses, a second, distinct out-of-distribution mode (`pfam_domain_check.json`).
- **FreeSASA:** burial correlates with predicted $|\Delta\Delta G|$ across the 195 2VUK positions (Spearman $\rho = 0.391$, $p = 1.6 \times 10^{-8}$): the magnitude signal is structurally sensible (`burial_analysis_2vuk.json`).
- **AlphaFold DB:** top SOD1 candidates sit at pLDDT ≈ 98.8 ; the model does not assign large effects to low-confidence regions (`alphafold_db_lookup.json`).
- **gnomAD:** TP53 is extremely loss-of-function-intolerant (pLI = 0.9996), consistent with germline depletion of core-domain destabilizing variants (`gnomad_tp53_constraint.json`).
- **STRING:** TP53/SOD1 partner landscape recorded for context (`string_network_context.json`).

8 Limitations and negative results

All of the following are preserved with quantification in `results/`: (1) v1 aromatic-identity bias (up to +1.04 kcal/mol for TRP); (2) greedy multi-mutant sums are implausible under the independence approximation (+19.5 v1, +46.0 v2); (3) p53 transfer collapses at long training (r 0.426 v1 $\rightarrow$ -0.02 v2-e300) and validation-based early stopping does not repair it; (4) proline positions are out of distribution because the training data contain no proline mutations; (5) SOD1 $\Delta\Delta G$ prediction fails outright ($r = -0.15$, $n = 8$); (6) the documented Y220C suppressor A138G is missed; (7) scan magnitudes are inflated and only ranks should be read; (8) the apparently strong MYOGLOBIN result ($r = 0.455$) is contaminated by 41 same-protein training rows and drops to 0.24-0.29 when decontaminated.

9 External tools

Scientific libraries and data services used include NumPy, SciPy, scikit-learn, Biopython, FreeSASA, RCSB PDB, FireProtDB, ThermoMutDB, UniProt and PDBe/SIFTS. The complete historical resource inventory and evidence links are in `results/EXTERNAL_TOOLS.md`; its 44 rows also include infrastructure, datasets, a research paper, literature-only lookups and a repeated RCSB endpoint, and therefore do not pass the strict 40-tool gate. The audited grouping is in Appendix A. Staged or planned tools are excluded. A new MaveDB SOD1 cross-check adds one genuine data API: the GNN scan has near-zero rank association with independent variant-abundance scores ($\rho = 0.032$, $n = 2,566$) and activity scores ($\rho = 0.043$, $n = 2,571$) after accounting for the mature-to-precursor residue offset (`results/mavedb_sod1_crosscheck.json`). These are functional readouts, not direct $\Delta\Delta G$ measurements, so they corroborate a model limitation without inflating the stability benchmark.

Appendix A: Audited scientific resources (strict 40-tool gate open)

This grouped list identifies verified uses and their result ledgers, not forty independent tool rows. Staged and planned uses are excluded.

- Scientific libraries and analysis: NumPy, SciPy, scikit-learn, pandas, Biopython, FreeSASA, matplotlib and PyMOL (models, structural verification, statistical testing and figures).
- Experimental and disease-data services: RCSB PDB, FireProtDB, ThermoMutDB, UniProt, PDBe SIFTS, AlphaFold DB, gnomAD, ClinVar, dbSNP, Open Targets, Ensembl, Reactome, STRING and PDB-REDO (the result files and accessions are in results/EXTERNAL_TOOLS.md).
- Structure/function services: Pfam through InterPro, CATH-Gene3D, 3D-Beacons, MobiDB, EBI Proteins, NCBI BLAST, InterProScan, Clustal Omega and EMBOSS pepstats. Repeated endpoint brands are not counted as new tools.

The earlier 44-row historical ledger is preserved in results/EXTERNAL_TOOLS.md, with status and evidence for each entry. A paper-only benchmark reference, datasets, specifications, infrastructure, and repeated endpoints do not qualify toward forty independently used scientific tools.

Strict-count audit. This historical table contains pytest, git/GitHub, Drive, TeX, a research paper, a specification, a dataset and literature lookup; none of those qualify as a distinct used scientific tool. The later results/EXTERNAL_TOOLS.md ledger adds further scientific services, but repeats RCSB under a second endpoint. Removing infrastructure, literature-only entries, dataset-only resources and the repeated RCSB endpoint leaves at most 34 eligible entries in the expanded 44-row ledger, before validating all evidence. MaveDB adds one newly executed research/data API with two separately archived SOD1 scoresets; the strict 40-tool gate remains open.

Appendix B: Per-identity bias, v1 vs v2 (S2648)

Mean residual (predicted – experimental $\Delta\Delta G$, kcal/mol) by mutant amino-acid identity. The v1 model systematically over-credits aromatic/hydrophobic identities; the v2 center-aware readout suppresses this by an order of magnitude.

Mutant identity	n (S2648)	v1 mean resid	v2 mean resid
ALA	744	+0.195	-0.160
ARG	51	+0.459	+0.127
ASN	73	+0.354	+0.062
ASP	74	+0.327	-0.022
CYS	66	+0.604	-0.072
GLN	89	+0.086	+0.127
GLU	66	+0.297	-0.021
GLY	307	+0.368	-0.075
HIS	50	+0.603	-0.003
ILE	118	+1.001	+0.191
LEU	155	+0.656	-0.006
LYS	70	+0.257	+0.016
MET	51	+0.501	-0.070

Mutant identity	n (S2648)	v1 mean resid	v2 mean resid
PHE	182	+0.602	-0.076
SER	129	+0.535	-0.005
THR	106	+0.805	-0.029
TRP	40	+1.038	-0.061
TYR	52	+0.882	+0.042
VAL	221	+0.877	+0.073

Appendix C: Top novel 2VUK rescue candidates (bias-corrected, PRO-masked)

Full ranking in build/scan_2vuk_whole_v2.json; top 250 in results/scan_candidates_2vuk_top250.csv. Ranks, not magnitudes, are the output.

Rank	Mutation	$\Delta\Delta G_{\text{pred}}$	$\Delta\Delta G_{\text{corr}}$
1	SER240CYS	+6.452	+6.524
2	GLY245CYS	+5.981	+6.052
3	MET246CYS	+5.947	+6.019
4	ARG273CYS	+5.842	+5.913
5	SER241CYS	+5.527	+5.599
6	ALA189ARG	+5.608	+5.481
7	MET243CYS	+5.153	+5.224
8	ALA161GLU	+5.148	+5.169
9	ALA161TYR	+5.189	+5.147
10	ASN247CYS	+5.039	+5.110
11	ALA138GLU	+5.061	+5.083
12	ALA138CYS	+4.973	+5.045
13	ALA161ARG	+5.059	+4.931
14	THR123CYS	+4.806	+4.878
15	TYR239CYS	+4.346	+4.418
16	ARG110ILE	+4.509	+4.318
17	GLY244CYS	+4.244	+4.316
18	ALA189TYR	+4.332	+4.290
19	SER215LEU	+4.231	+4.237
20	GLY279MET	+4.153	+4.223
21	ALA161HIS	+4.197	+4.200
22	ALA129HIS	+4.172	+4.175
23	ALA276CYS	+4.103	+4.175
24	HIS178CYS	+4.080	+4.152
25	ALA138MET	+4.077	+4.147

Appendix E: Top WT-SOD1 (1SPD) stabilizing candidates

Hypotheses only - the model does not validate on SOD1 (Sec. 5.6); listed so future work can test or refute them.

Rank	Mutation	$\Delta\Delta G_{\text{pred}}$	$\Delta\Delta G_{\text{corr}}$
1	GLY82ASN	+5.878	+5.816
2	GLY33TYR	+5.738	+5.696
3	GLY33TRP	+5.277	+5.338
4	GLY82LYS	+5.157	+5.141
5	GLY41TRP	+5.001	+5.062
6	GLY82HIS	+4.932	+4.935
7	GLY41ILE	+5.044	+4.853
8	GLY82LEU	+4.824	+4.830
9	ALA4TYR	+4.866	+4.824
10	GLY82ARG	+4.756	+4.629
11	GLY41PHE	+4.385	+4.460
12	ALA4TRP	+4.314	+4.375
13	GLY147CYS	+4.223	+4.295
14	ASN19TYR	+4.214	+4.172
15	ALA145TYR	+4.186	+4.144
16	GLY82THR	+4.109	+4.139
17	ALA145CYS	+4.058	+4.130
18	ALA4LEU	+4.098	+4.103
19	ALA4CYS	+3.976	+4.048
20	GLY127LEU	+4.007	+4.013
21	GLY82ILE	+4.176	+3.986
22	ALA60ARG	+4.112	+3.985
23	GLY33PHE	+3.905	+3.981
24	GLY16CYS	+3.904	+3.976
25	THR137TRP	+3.868	+3.929

Appendix F: Reproducibility

Environment

Python 3.10, NumPy, SciPy, scikit-learn, pandas, Biopython, FreeSASA 2.2.1, PyMOL open-source, matplotlib, TeX Live (mathptmx). CPU-only; no GPU. Sandbox: 2 CPU / 2 GB RAM.

Exact commands

- Training (benchmark model): `python3 scripts/train_bench.py --arch v2 --epochs 300 --hidden 64 --mlp-hidden 32 --lr 0.001 --batch 32 --seed 0 --ckpt build/ckpt_v2.pkl --out build/bench_v2_e300.json`
- v1 model: same with `--arch v1 --epochs 150 --hidden 32 --mlp-hidden 16 --lr 0.0005`.
- Validation-split study: `--val-frac 0.15 --val-every 5` (PDB-grouped split, seed 0).
- Bias: `scripts/bias_measure.py`; scans: `scripts/rescue_scan_v2_whole.py` (2VUK, 1SPD); validation: `scripts/sod1_validate.py`; statistics: `scripts/stats_and_baselines.py` (5000-resample bootstrap, seed fixed); figures: `scripts/make_figures.py`.
- Tool CLI: `tools/gnn-ddg scan --pdb 2VUK --chain A`.

- Tests: `python3 -m pytest tests/` (27 hermetic tests, no network).

Leakage protocol

S2648/Ssym share no exact (structure, position, WT, mutant) rows on the inverse split (verified: 0/342 overlap, committed in `results/ssym_benchmark_v2_e300.json`). Direct-split overlap with published training sets is expected and is why the inverse split is the reported benchmark. Checkpoints, predictions and every quoted number regenerate deterministically from the committed scripts with the seeds above; the altloc parser bug and its superseded artifacts are preserved in `build/superseded_altloc/` as an audit trail.

Appendix D: Identified source records (481 PDB accessions + 12 resources)

Inventory of separately identified source records: 481 unique PDB accessions plus 12 named resources. This is not 493 independent experimental studies. Provenance per entry: `results/DATASETS.md`.

Named datasets (12): ProtDDG-Bench S2648; Ssym direct; Ssym inverse; P53; Pucci et al. 2018 Table 1; ThermoMutDB human SOD1; FireProtDB human SOD1; known p53 Y220C suppressor set; SoDCoD SOD1; UniProt P00441; UniProt P04637; Zenodo 8169289. These are named resources, not all independent study datasets.

Distinct PDB accessions (481)

107L	108L	109L	110L
111L	112L	113L	114L
115L	118L	119L	120L
122L	123L	125L	126L
127L	128L	137L	149L
150L	172L	1A43	1A5E
1AEP	1AG2	1AJ3	1AKY
1AM7	1AMQ	1ANK	1AON
1APS	1ARR	1AZP	1B26
1B7S	1B8E	1BAN	1BAO
1BLC	1BNI	1BNS	1BOY
1BPT	1BRG	1BRH	1BRI
1BRJ	1BRK	1BSA	1BSB
1BSC	1BSD	1BSE	1BTA
1BTI	1BVC	1C9O	1CAH
1CEY	1CHK	1CSE	1CSP
1CTW	1CU0	1CU2	1CU3
1CU5	1CU6	1CUN	1CUP
1CUQ	1CV0	1CV1	1CV3
1CV4	1CV5	1CV6	1CVK
1D2W	1D2Y	1D3J	1DKT
1DMM	1DMQ	1DYA	1DYB
1DYC	1DYD	1DYE	1DYF
1DYG	1E65	1E6K	1E6L
1E6M	1EY0	1EY4	1EY5
1EY6	1EY7	1FAN	1FKJ
1FLQ	1FLU	1FLW	1FLY

1FN5	1FNA	1FTG	1FVK
1G06	1G07	1G0G	1G0J
1G0K	1G0L	1G0M	1G0P
1G0Q	1G1V	1G1W	1G4I
1G6N	1GAY	1GB0	1GB2
1GB3	1GB5	1GB6	1GB7
1GB8	1GB9	1GB0	1GBW
1GBX	1GBY	1GBZ	1GF8
1GF9	1GFA	1GFE	1GFG
1GFH	1GFJ	1GFK	1GFR
1GFT	1GFU	1GFV	1GOB
1H7M	1HEM	1HEO	1HER
1HFZ	1HIB	1HK0	1HME
1HMK	1HMS	1HNL	1HTI
1HUU	1I6S	1IET	1IFC
1IGV	1IHB	1IMQ	1INU
1IO2	1IOB	1IOS	1IOT
1IR7	1IR8	1IR9	1IRO
1JIW	1K9Q	1KAB	1KCQ
1KDX	1KE4	1KFW	1KVA
1KVB	1KVC	1L00	1L02
1L03	1L04	1L06	1L07
1L08	1L09	1L10	1L11
1L12	1L13	1L14	1L15
1L16	1L17	1L18	1L19
1L20	1L21	1L22	1L23
1L24	1L33	1L37	1L38
1L39	1L42	1L44	1L45
1L46	1L47	1L48	1L52
1L53	1L55	1L56	1L57
1L59	1L60	1L61	1L62
1L63	1L65	1L66	1L67
1L68	1L69	1L76	1L77
1L85	1L86	1L87	1L88
1L90	1L91	1L92	1L93
1L94	1L95	1L96	1L98
1L99	1LAV	1LAW	1LBI
1LHH	1LHI	1LHJ	1LHK
1LHL	1LNI	1LSN	1LUC
1LVE	1LYE	1LYF	1LYG
1LYH	1LYI	1LYJ	1LZ1
1MBG	1MGR	1MJC	1MSI
1MX2	1MX4	1MX6	1N0J
1NAG	1OH0	1OIA	1ONC
1OTR	1OUA	1OUB	1OUC
1OUD	1OUE	1OUF	1OUG
1OUH	1OUI	1OUJ	1P2L

1P2P	1P2R	1P36	1P46
1P64	1P6Y	1P7S	1PDO
1PGA	1POH	1QGV	1QIR
1QIS	1QIT	1QLP	1QM4
1QND	1QS5	1QS9	1QSB
1QT6	1QT7	1QTB	1QTC
1QTD	1RBR	1RBS	1RBT
1RBU	1RBV	1RDA	1RDB
1RDC	1RG8	1RGC	1RHG
1RIS	1RN1	1ROP	1RTB
1RTP	1SAK	1SHF	1SHG
1SPD	1SSO	1SUP	1SYC
1SYE	1SYG	1TEN	1TIT
1TLA	1TPK	1TSR	1TTQ
1TYV	1UBQ	1UZC	1VQB
1VQG	1VQH	1VQI	1VQJ
1WIT	1WQ5	1WQM	1WQN
1WQO	1WQP	1WQQ	1WQR
1YAM	1YAN	1YAO	1YAP
1YAQ	1YHB	1YU5	1YYJ
1ZG4	1ZNJ	206L	217L
221L	224L	230L	231L
232L	233L	234L	235L
236L	237L	238L	239L
240L	241L	242L	243L
244L	245L	246L	247L
253L	254L	255L	2A01
2A36	2ABD	2C9V	2CI2
2DRI	2EXZ	2EY1	2EY2
2EY5	2EY6	2EYF	2EYH
2EYJ	2EYL	2EYM	2EYO
2EYP	2F0D	2F0E	2F0F
2F0G	2F0H	2F0I	2F0J
2H61	2HEA	2HEB	2HEC
2HED	2HEE	2HEF	2HPR
2IMM	2J1X	2L78	2LZM
2NVH	2OCJ	2RN2	2SNM
2TRT	2TRX	2TS1	2VUK
2XWR	3AA2	3AA3	3AA4
3AA5	3C7W	3C7Y	3C80
3C81	3C83	3C8Q	3C8R
3C8S	3CDO	3CDQ	3CDR
3CDT	3CDV	3ECA	3F8V
3FI5	3GLY	3HHR	3MBP
3PGK	3SIL	3SSI	4LYZ
5CRO	5DFR	5EAA	5PTI

Appendix G: Ssym inverse benchmark, per-protein breakdown

Every one of the 342 proteins in the leak-free inverse test set, sorted by per-protein MAE of v2-e300. All values computed by `scripts/error_analysis.py`; machine-readable source: `build/ssym_inv_per_protein.json`. Median per-protein MAE 0.578, mean 0.958 kcal/mol; the mean is inflated by a small tail of hard proteins, visible at the bottom of the table.

#	PDB	n mutations	MAE (kcal/mol)
1	1GAY	1	0.006
2	2EYH	1	0.013
3	1LAV	1	0.015
4	1L65	1	0.015
5	3AA2	1	0.017
6	1BSB	1	0.017
7	1DYF	1	0.017
8	128L	1	0.019
9	1L23	1	0.020
10	1MX4	1	0.022
11	1E6K	1	0.027
12	1OUG	1	0.040
13	1L44	1	0.043
14	109L	1	0.050
15	1L38	1	0.050
16	1GBX	1	0.052
17	1BAN	1	0.057
18	1FLW	1	0.058
19	2EY1	1	0.059
20	2HEB	1	0.059
21	1WQN	1	0.066
22	1EY7	1	0.070
23	253L	1	0.071
24	1GF9	1	0.074
25	1L06	1	0.077
26	1IOT	1	0.078
27	3AA5	1	0.082
28	2EYF	1	0.082
29	1G0P	1	0.084
30	1RBV	1	0.087
31	2F0G	1	0.088
32	1MX2	1	0.093
33	1CV5	1	0.094
34	1HER	1	0.098
35	1RDA	1	0.104
36	1WQP	1	0.106

#	PDB	n mutations	MAE (kcal/mol)
37	238L	1	0.108
38	1SYG	1	0.110
39	1DMM	1	0.117
40	2EYL	1	0.117
41	1GFG	1	0.122
42	1KVC	1	0.128
43	1CV3	1	0.130
44	1GFE	1	0.130
45	1RBS	1	0.131
46	1L19	1	0.140
47	2F0J	1	0.141
48	1BNS	1	0.142
49	1G0G	1	0.145
50	1RBU	1	0.147
51	1RDB	1	0.152
52	2EY2	1	0.152
53	1DMQ	1	0.159
54	1GBO	1	0.160
55	1CV4	1	0.164
56	1D3J	1	0.173
57	1L66	1	0.175
58	115L	1	0.176
59	1GBW	1	0.177
60	1L14	1	0.180
61	1G06	1	0.192
62	1OUJ	1	0.192
63	1L13	1	0.193
64	2EYM	1	0.196
65	1L77	1	0.198
66	1GFH	1	0.200
67	1L22	1	0.218
68	1L20	1	0.227
69	119L	1	0.229
70	1WQR	1	0.230
71	1YAO	1	0.231
72	1IR7	1	0.239
73	1P64	1	0.240
74	1KVA	1	0.245
75	1GFK	1	0.246
76	1GBY	1	0.246
77	2EY6	1	0.247
78	1QT7	1	0.250
79	1L03	1	0.253
80	1FN5	1	0.253
81	1INU	1	0.258
82	1DYG	1	0.261
83	1GFT	1	0.261

#	PDB	n mutations	MAE (kcal/mol)
84	235L	1	0.266
85	1L10	1	0.278
86	137L	1	0.284
87	1L60	1	0.285
88	1OUC	1	0.286
89	1BAO	1	0.294
90	2EYP	1	0.295
91	1L45	1	0.295
92	3C81	1	0.309
93	1L92	1	0.313
94	1L53	1	0.314
95	1YAQ	1	0.314
96	1P2R	1	0.316
97	1L11	1	0.318
98	224L	1	0.321
99	122L	1	0.324
100	1BTI	1	0.333
101	2F0D	1	0.337
102	1QT6	1	0.340
103	1L52	1	0.340
104	1P2L	1	0.342
105	1DYB	1	0.345
106	1GB2	1	0.346
107	1KVB	1	0.351
108	1CU5	1	0.351
109	1GB3	1	0.352
110	1WQO	1	0.357
111	3C7Y	1	0.365
112	3AA4	1	0.366
113	1BSC	1	0.367
114	2F0H	1	0.370
115	1VQJ	1	0.381
116	1L09	1	0.382
117	1GB0	1	0.388
118	1DYC	1	0.391
119	1G0L	1	0.395
120	1SYC	1	0.396
121	1GFV	1	0.398
122	1CTW	1	0.400
123	217L	1	0.401
124	1EY4	1	0.403
125	1DYA	1	0.406
126	1L88	1	0.410
127	2F0E	1	0.414
128	1WQQ	1	0.414
129	1IR8	1	0.414
130	1L02	1	0.417

#	PDB	n mutations	MAE (kcal/mol)
131	1YAN	1	0.418
132	245L	1	0.430
133	1VQI	1	0.436
134	1L18	1	0.436
135	1L04	1	0.438
136	1GFU	1	0.441
137	1RDC	1	0.444
138	1OUE	1	0.444
139	1OUD	1	0.449
140	1GFA	1	0.451
141	1L08	1	0.459
142	1DYE	1	0.470
143	1L21	1	0.476
144	1GBZ	1	0.478
145	1L94	1	0.480
146	1HEM	1	0.483
147	1G07	1	0.489
148	1BRK	1	0.493
149	1MX6	1	0.494
150	1FLQ	1	0.499
151	1GOB	1	0.500
152	237L	1	0.503
153	2F0F	1	0.505
154	1L61	1	0.508
155	1LAW	1	0.515
156	1OUH	1	0.515
157	1G0J	1	0.521
158	1WQM	1	0.522
159	2F0I	1	0.527
160	1L37	1	0.530
161	1FLU	1	0.538
162	118L	1	0.541
163	1L17	1	0.543
164	1VQG	1	0.552
165	1BRH	1	0.553
166	3C8Q	1	0.560
167	1G0M	1	0.562
168	1L15	1	0.568
169	1DYG	1	0.574
170	1GF8	1	0.575
171	3F8V	1	0.578
172	1GB8	1	0.578
173	1L16	1	0.591
174	127L	1	0.593
175	1GB7	1	0.602
176	1YAM	1	0.605
177	123L	1	0.607

#	PDB	n mutations	MAE (kcal/mol)
178	1L12	1	0.611
179	111L	1	0.615
180	107L	1	0.622
181	1L67	1	0.622
182	1G1W	1	0.625
183	244L	1	0.626
184	1L59	1	0.627
185	126L	1	0.634
186	1KAB	1	0.637
187	1GB5	1	0.637
188	1IR9	1	0.642
189	1LSN	1	0.649
190	113L	1	0.650
191	1TLA	1	0.652
192	2EYO	1	0.660
193	3CDQ	1	0.660
194	1FLY	1	0.672
195	1L47	1	0.673
196	1RGC	1	0.684
197	1L46	1	0.692
198	2HEE	1	0.692
199	1BRI	1	0.703
200	1EY5	1	0.710
201	1L62	1	0.715
202	1OUF	1	0.716
203	1L87	1	0.720
204	1BSE	1	0.731
205	1BSD	1	0.732
206	1QIR	1	0.740
207	1BSA	1	0.746
208	2EY5	1	0.751
209	2EXZ	1	0.753
210	1L69	1	0.760
211	3C83	1	0.761
212	1QTC	1	0.764
213	149L	1	0.767
214	1GFJ	1	0.821
215	2HEF	1	0.821
216	1L68	1	0.829
217	1CUQ	1	0.838
218	108L	1	0.840
219	1RBT	1	0.843
220	241L	1	0.845
221	1L39	1	0.846
222	120L	1	0.855
223	1NAG	1	0.861
224	1L98	1	0.865

#	PDB	n mutations	MAE (kcal/mol)
225	1OUA	1	0.871
226	1VQH	1	0.879
227	1GB6	1	0.880
228	232L	1	0.909
229	112L	1	0.918
230	1L57	1	0.918
231	2HEC	1	0.938
232	1OUB	1	0.940
233	240L	1	0.943
234	150L	1	0.949
235	1E6L	1	0.966
236	1B7S	1	0.972
237	5EAA	1	0.991
238	1L42	1	1.000
239	1YAP	1	1.005
240	3C8R	1	1.010
241	239L	1	1.012
242	3AA3	1	1.034
243	1L86	1	1.063
244	2HED	1	1.067
245	1L91	1	1.074
246	1BRJ	1	1.078
247	1EY6	1	1.087
248	1L90	1	1.101
249	1GFR	1	1.112
250	1HEO	1	1.117
251	1G1V	1	1.134
252	221L	1	1.136
253	1L33	1	1.141
254	110L	1	1.174
255	1GB9	1	1.177
256	3CDT	1	1.179
257	1SYE	1	1.210
258	1LHL	1	1.221
259	1HIB	1	1.234
260	1LHK	1	1.235
261	1L07	1	1.255
262	1P7S	1	1.286
263	1L93	1	1.289
264	1OUI	1	1.303
265	3C8S	1	1.327
266	114L	1	1.371
267	2HEA	1	1.374
268	1IOS	1	1.405
269	3C7W	1	1.412
270	1G0K	1	1.414
271	243L	1	1.416

#	PDB	n mutations	MAE (kcal/mol)
272	1L48	1	1.440
273	125L	1	1.449
274	236L	1	1.502
275	1L55	1	1.504
276	247L	1	1.511
277	231L	1	1.513
278	2EYJ	1	1.528
279	1E6M	1	1.531
280	1CV1	1	1.568
281	1QIS	1	1.648
282	1CVK	1	1.685
283	1HNL	1	1.689
284	1L85	1	1.724
285	2L78	1	1.738
286	1BPT	1	1.757
287	1CUP	1	1.794
288	1LYI	1	1.846
289	242L	1	1.864
290	1CV0	1	1.913
291	1QS9	1	1.993
292	8PTI	1	1.999
293	1QSB	1	2.004
294	1LHI	1	2.010
295	3CDR	1	2.036
296	254L	1	2.038
297	1L00	1	2.072
298	1QTD	1	2.079
299	1CU2	1	2.092
300	233L	1	2.146
301	1D2Y	1	2.154
302	1CU0	1	2.209
303	206L	1	2.227
304	1CU6	1	2.232
305	1G0Q	1	2.300
306	1I6S	1	2.342
307	1CV6	1	2.387
308	255L	1	2.410
309	1P36	1	2.414
310	1LYH	1	2.426
311	1L76	1	2.563
312	1P6Y	1	2.585
313	1L99	1	2.667
314	1QIT	1	2.738
315	246L	1	2.792
316	1LYF	1	2.839
317	3CDO	1	2.892
318	1CU3	1	2.900

#	PDB	n mutations	MAE (kcal/mol)
319	1D2W	1	2.903
320	1BRG	1	2.941
321	1P46	1	2.942
322	1LYG	1	2.960
323	172L	1	3.004
324	234L	1	3.032
325	1L95	1	3.068
326	1L24	1	3.151
327	1LHH	1	3.241
328	1LYJ	1	3.336
329	1YHB	1	3.396
330	3CDV	1	3.620
331	230L	1	3.805
332	1L56	1	3.864
333	2SNM	1	3.888
334	1FAN	1	3.934
335	1LYE	1	3.978
336	1RBR	1	4.042
337	3FI5	1	4.262
338	1QTB	1	4.374
339	3C80	1	4.900
340	1QS5	1	5.618
341	1LHJ	1	6.108
342	1L96	1	6.741

Appendix H: 2VUK whole-chain scan, top 600 (bias-corrected)

Full single-mutant scan of 2VUK chain A (195 positions, 19 mutants each, 3705 rows; 266 WT=PRO rows out-of-distribution and excluded here), ranked by $|\Delta\Delta G_{\text{corr}}|$. *Refs* marks mutants present in the S2648/Ssym reference sets for this protein (leakage-checked). Ranks, not magnitudes, are the output. Source: build/scan_2vuk_whole_ (generator scripts/scan_whole.py).

#	Mutation	$\Delta\Delta G_p$	$\Delta\Delta G_c$	Refs
1	ARG158PRO	-9.473	-9.473	
2	ILE162GLY	-8.743	-8.668	
3	ILE255PRO	-8.438	-8.438	
4	ILE162PRO	-8.225	-8.225	
5	TYR126PRO	-8.220	-8.220	
6	TYR205PRO	-8.143	-8.143	
7	ARG175PRO	-7.803	-7.803	
8	ARG196PRO	-7.643	-7.643	
9	VAL197PRO	-7.376	-7.376	
10	VAL217PRO	-7.355	-7.355	
11	VAL272PRO	-7.297	-7.297	
12	SER215PRO	-7.179	-7.179	
13	GLU204PRO	-7.107	-7.107	
14	VAL157PRO	-7.038	-7.038	
15	CYS124PRO	-6.997	-6.997	

#	Mutation	$\Delta\Delta G_p$	$\Delta\Delta G_c$	Refs
16	CYS176PRO	-6.946	-6.946	
17	CYS238PRO	-6.944	-6.944	
18	ARG174PRO	-6.939	-6.939	
19	VAL216PRO	-6.905	-6.905	
20	TYR234PRO	-6.875	-6.875	
21	TYR236PRO	-6.795	-6.795	
22	CYS176ILE	-6.563	-6.754	
23	CYS176TRP	-6.806	-6.745	
24	ILE162ALA	-6.865	-6.705	
25	ARG175ALA	-6.798	-6.638	
26	ILE195PRO	-6.596	-6.596	
27	THR256PRO	-6.532	-6.532	
28	SER240CYS	+6.452	+6.524	
29	CYS242ILE	-6.323	-6.514	
30	TYR163PRO	-6.428	-6.428	
31	LEU206PRO	-6.294	-6.294	
32	CYS176ALA	-6.404	-6.244	
33	MET160PRO	-6.186	-6.186	
34	CYS238ALA	-6.343	-6.184	
35	GLN144PRO	-6.180	-6.180	
36	GLY245CYS	+5.981	+6.052	
37	MET246CYS	+5.947	+6.019	
38	ARG174ALA	-6.160	-6.001	
39	ARG202PRO	-5.958	-5.958	
40	ARG181ALA	-6.106	-5.946	
41	ARG273CYS	+5.842	+5.913	
42	HIS214PRO	-5.884	-5.884	
43	LEU194PRO	-5.878	-5.878	
44	CYS242PRO	-5.861	-5.861	
45	GLU271PRO	-5.822	-5.822	
46	ARG213PRO	-5.804	-5.804	
47	ARG181PRO	-5.724	-5.724	
48	GLU198PRO	-5.716	-5.716	
49	CYS275PRO	-5.711	-5.711	
50	LEU201PRO	-5.628	-5.628	
51	LEU252PRO	-5.620	-5.620	
52	SER241CYS	+5.527	+5.599	
53	ASN200PRO	-5.561	-5.561	
54	CYS242ALA	-5.713	-5.553	
55	VAL173PRO	-5.538	-5.538	
56	CYS242VAL	-5.421	-5.494	
57	ALA189ARG	+5.608	+5.481	
58	ILE251PRO	-5.480	-5.480	
59	VAL172PRO	-5.475	-5.475	
60	VAL218PRO	-5.466	-5.466	
61	ARG181HIS	-5.451	-5.448	
62	ARG282PRO	-5.433	-5.433	
63	LEU145PRO	-5.418	-5.418	
64	LYS132PRO	-5.412	-5.412	
65	PHE113PRO	-5.319	-5.319	
66	PHE134PRO	-5.269	-5.269	
67	ARG273PRO	-5.261	-5.261	
68	CYS135PRO	-5.252	-5.252	

#	Mutation	$\Delta\Delta G_p$	$\Delta\Delta G_c$	Refs
69	MET243CYS	+5.153	+5.224	
70	ARG175ILE	-4.987	-5.178	
71	ALA161GLU	+5.148	+5.169	
72	ARG175TRP	-5.221	-5.160	
73	ALA161TYR	+5.189	+5.147	
74	CYS124ASN	-5.066	-5.128	
75	ASN247CYS	+5.039	+5.110	
76	ILE162SER	-5.097	-5.092	
77	ALA138GLU	+5.061	+5.083	
78	CYS242TRP	-5.135	-5.074	
79	ILE162HIS	-5.067	-5.064	
80	CYS176VAL	-4.973	-5.046	
81	ALA138CYS	+4.973	+5.045	
82	TYR126GLY	-5.112	-5.037	
83	CYS141PRO	-5.031	-5.031	
84	ARG156PRO	-4.988	-4.988	
85	HIS193PRO	-4.968	-4.968	
86	ARG174GLY	-5.028	-4.953	
87	ARG175MET	-5.017	-4.947	
88	ALA161ARG	+5.059	+4.931	
89	TYR239PRO	-4.907	-4.907	
90	ARG181MET	-4.974	-4.905	
91	ARG181PHE	-4.976	-4.900	
92	VAL122ALA	-5.040	-4.880	
93	CYS242GLY	-4.954	-4.879	
94	THR123CYS	+4.806	+4.878	
95	ARG282GLY	-4.953	-4.878	
96	ILE162ASP	-4.875	-4.853	
97	LEU252GLY	-4.925	-4.850	
98	THR253PRO	-4.819	-4.819	
99	ALA159PRO	-4.816	-4.816	
100	ASP281PRO	-4.792	-4.792	
101	CYS238TRP	-4.853	-4.792	
102	CYS238ILE	-4.575	-4.766	
103	ARG181THR	-4.794	-4.764	
104	LEU111PRO	-4.763	-4.763	
105	ILE162GLN	-4.634	-4.761	
106	CYS176MET	-4.802	-4.733	
107	ALA203PRO	-4.727	-4.727	
108	CYS135ALA	-4.885	-4.725	
109	ARG282ALA	-4.875	-4.715	
110	GLU258PRO	-4.697	-4.697	
111	GLU285GLY	-4.766	-4.691	
112	ILE255LYS	-4.675	-4.691	
113	ILE162ASN	-4.612	-4.674	
114	TYR163GLY	-4.729	-4.653	
115	CYS238GLY	-4.713	-4.637	
116	PHE270PRO	-4.610	-4.610	
117	CYS124LYS	-4.590	-4.606	
118	LYS164PRO	-4.596	-4.596	
119	CYS176GLN	-4.452	-4.579	
120	CYS124ALA	-4.723	-4.563	
121	LEU257PRO	-4.547	-4.547	

#	Mutation	$\Delta\Delta G_p$	$\Delta\Delta G_c$	Refs
122	THR231PRO	-4.538	-4.538	
123	VAL143PRO	-4.529	-4.529	
124	GLN136PRO	-4.522	-4.522	
125	GLY112PRO	-4.497	-4.497	
126	CYS176HIS	-4.465	-4.462	
127	CYS242ASN	-4.399	-4.460	
128	CYS124SER	-4.453	-4.448	
129	TYR239CYS	+4.346	+4.418	
130	ARG175GLY	-4.482	-4.407	
131	CYS275ALA	-4.529	-4.369	
132	ILE251ALA	-4.498	-4.338	
133	ARG110ILE	+4.509	+4.318	
134	GLY244CYS	+4.244	+4.316	
135	ARG175VAL	-4.227	-4.300	
136	THR230PRO	-4.300	-4.300	
137	ARG248PRO	-4.300	-4.300	
138	CYS242HIS	-4.301	-4.298	
139	CYS135SER	-4.299	-4.294	
140	ARG181GLY	-4.366	-4.291	
141	THR125PRO	-4.291	-4.291	
142	CYS176ASN	-4.229	-4.290	
143	ALA189TYR	+4.332	+4.290	
144	ASP281ALA	-4.444	-4.284	
145	ILE251GLY	-4.320	-4.245	
146	SER215LEU	+4.231	+4.237	
147	ILE162MET	-4.304	-4.234	
148	CYS238SER	-4.228	-4.223	
149	GLY279MET	+4.153	+4.223	
150	ASP281GLY	-4.288	-4.213	
151	ARG181ILE	-4.018	-4.208	
152	ALA161HIS	+4.197	+4.200	
153	CYS176GLY	-4.266	-4.191	
154	THR155PRO	-4.182	-4.182	
155	LEU133PRO	-4.178	-4.178	
156	ALA129HIS	+4.172	+4.175	
157	ALA276CYS	+4.103	+4.175	
158	ARG267PRO	-4.174	-4.174	
159	HIS178CYS	+4.080	+4.152	
160	ALA138MET	+4.077	+4.147	
161	CYS176GLU	-4.162	-4.141	
162	ARG181ASP	-4.161	-4.138	
163	ARG175SER	-4.140	-4.136	
164	VAL172GLY	-4.190	-4.115	
165	LYS139CYS	+4.043	+4.114	
166	CYS176SER	-4.116	-4.111	
167	CYS176PHE	-4.176	-4.100	
168	HIS179CYS	+4.026	+4.098	
169	LEU137HIS	+4.068	+4.071	
170	ILE255SER	-4.058	-4.053	
171	ASN235PRO	-4.047	-4.047	
172	ARG175HIS	-4.047	-4.044	
173	CYS242MET	-4.099	-4.030	
174	GLU285ALA	-4.186	-4.026	

#	Mutation	$\Delta\Delta G_p$	$\Delta\Delta G_c$	Refs
175	CYS242GLN	-3.890	-4.017	
176	ALA161LEU	+4.009	+4.015	
177	LEU114ARG	+4.136	+4.009	
178	ARG181SER	-4.013	-4.008	
179	ILE255GLY	-4.081	-4.006	
180	LEU194ALA	-4.138	-3.978	
181	CYS124GLN	-3.845	-3.972	
182	CYS238VAL	-3.899	-3.972	
183	LEU252ALA	-4.131	-3.971	
184	GLN192PRO	-3.965	-3.965	
185	CYS220PRO	-3.961	-3.961	
186	LEU188PRO	-3.960	-3.960	
187	CYS176LEU	-3.965	-3.959	
188	MET237PRO	-3.949	-3.949	
189	CYS242LEU	-3.953	-3.947	
190	CYS238MET	-4.013	-3.943	
191	GLU285LEU	-3.939	-3.933	
192	ARG181VAL	-3.848	-3.921	
193	THR211CYS	+3.834	+3.906	
194	ASP207PRO	-3.902	-3.902	
195	VAL122PRO	-3.895	-3.895	
196	ARG213GLN	-3.747	-3.873	
197	THR123PRO	-3.873	-3.873	yes
198	ARG249PRO	-3.836	-3.836	
199	GLU180CYS	+3.763	+3.835	
200	SER183HIS	-3.834	-3.830	
201	ILE162LYS	-3.806	-3.822	
202	ARG181TRP	-3.881	-3.821	
203	CYS275ASN	-3.756	-3.817	
204	ARG181LEU	-3.806	-3.800	
205	ALA138PHE	+3.724	+3.799	
206	ARG248ALA	-3.952	-3.792	
207	CYS275SER	-3.793	-3.788	
208	ALA119TRP	+3.709	+3.770	
209	ALA161THR	+3.714	+3.744	
210	LEU137CYS	+3.672	+3.744	
211	ILE162THR	-3.765	-3.736	
212	VAL173GLY	-3.808	-3.733	
213	VAL122SER	-3.730	-3.725	
214	LYS139LEU	+3.717	+3.723	
215	CYS277PRO	-3.720	-3.720	
216	VAL122ASN	-3.646	-3.707	
217	ARG175ASP	-3.714	-3.692	
218	LYS139GLU	+3.670	+3.691	
219	SER127CYS	+3.592	+3.664	
220	ARG175LEU	-3.668	-3.662	
221	CYS242THR	-3.691	-3.661	
222	ALA161PHE	+3.584	+3.660	
223	CYS124GLY	-3.732	-3.657	
224	ALA159GLU	+3.634	+3.655	
225	VAL122LYS	-3.626	-3.642	
226	CYS275LYS	-3.622	-3.638	
227	GLY279VAL	+3.707	+3.635	

#	Mutation	$\Delta\Delta G_p$	$\Delta\Delta G_c$	Refs
228	ASP268PRO	-3.626	-3.626	
229	GLY108TRP	+3.557	+3.618	
230	ILE162PHE	-3.677	-3.602	
231	HIS168VAL	+3.673	+3.600	
232	PHE134SER	-3.599	-3.595	
233	ARG110PRO	-3.594	-3.594	
234	TYR163ALA	-3.752	-3.592	
235	GLU171GLY	-3.663	-3.588	
236	CYS242ARG	-3.458	-3.585	
237	GLY279CYS	+3.512	+3.584	
238	ALA203ARG	+3.708	+3.581	
239	VAL173ALA	-3.739	-3.579	
240	ALA159TYR	+3.613	+3.571	
241	ILE232PRO	-3.570	-3.570	
242	VAL272GLY	-3.642	-3.567	
243	ARG174GLN	-3.434	-3.561	
244	ARG175GLN	-3.434	-3.560	
245	HIS193LEU	+3.549	+3.555	
246	LEU137GLU	+3.516	+3.537	
247	LEU252ASN	-3.466	-3.528	
248	ALA138LEU	+3.521	+3.527	
249	LEU114TYR	+3.569	+3.527	
250	ARG174ILE	-3.327	-3.518	
251	ASP186PRO	-3.515	-3.515	
252	ARG174SER	-3.518	-3.513	
253	GLU286GLY	-3.577	-3.502	
254	ARG196ALA	-3.661	-3.501	
255	HIS233PRO	-3.501	-3.501	
256	VAL172ALA	-3.658	-3.498	
257	SER183LEU	-3.502	-3.496	
258	ALA189SER	+3.488	+3.493	
259	ALA159LEU	+3.484	+3.490	
260	GLY199PRO	-3.488	-3.488	
261	CYS135LYS	-3.467	-3.483	
262	CYS242SER	-3.486	-3.481	
263	THR211PRO	-3.474	-3.474	
264	LEU289MET	+3.403	+3.472	
265	ARG213MET	-3.529	-3.459	
266	CYS238LEU	-3.457	-3.451	
267	TYR236CYS	+3.377	+3.448	
268	GLN104PRO	-3.447	-3.447	
269	ARG174TRP	-3.502	-3.442	
270	ASP208PRO	-3.438	-3.438	
271	HIS168ARG	+3.551	+3.424	yes
272	CYS242GLU	-3.445	-3.424	
273	TYR103PRO	-3.419	-3.419	
274	ASP281ASN	-3.349	-3.410	
275	ARG175ASN	-3.343	-3.404	
276	VAL274CYS	+3.318	+3.390	
277	LEU252SER	-3.385	-3.380	
278	GLY117ILE	+3.571	+3.380	
279	ARG175LYS	-3.359	-3.375	
280	CYS176THR	-3.397	-3.368	

#	Mutation	$\Delta\Delta G_p$	$\Delta\Delta G_c$	Refs
281	PHE212CYS	+3.291	+3.363	yes
282	PHE134ALA	-3.522	-3.363	
283	VAL274PRO	-3.360	-3.360	
284	ARG248TRP	-3.418	-3.357	
285	ARG174MET	-3.425	-3.355	
286	CYS229PRO	-3.353	-3.353	
287	PHE134ASN	-3.275	-3.337	
288	LEU137ARG	+3.429	+3.301	
289	LYS139VAL	+3.372	+3.300	
290	CYS176ASP	-3.299	-3.277	
291	ARG273ILE	+3.467	+3.276	
292	TYR126SER	-3.274	-3.270	
293	SER149PRO	-3.268	-3.268	
294	VAL272ASP	-3.287	-3.265	
295	SER185PRO	-3.251	-3.251	
296	CYS275GLN	-3.121	-3.248	
297	ILE251SER	-3.247	-3.243	
298	ARG213GLY	-3.315	-3.240	
299	THR140LEU	+3.228	+3.234	
300	CYS242PHE	-3.310	-3.234	
301	ARG213ALA	-3.391	-3.231	
302	ALA138VAL	+3.301	+3.229	
303	ALA159ARG	+3.356	+3.229	
304	ARG181GLU	-3.249	-3.228	
305	ILE162TRP	-3.285	-3.225	
306	TRP146PRO	-3.218	-3.218	
307	GLN100PRO	-3.215	-3.215	
308	VAL147PRO	-3.210	-3.210	
309	ILE162ARG	-3.082	-3.209	
310	CYS176LYS	-3.182	-3.198	
311	HIS193ARG	+3.325	+3.198	
312	VAL225ARG	+3.323	+3.195	
313	ARG181LYS	-3.178	-3.194	
314	VAL122GLN	-3.064	-3.191	
315	SER185LEU	-3.194	-3.188	
316	ALA138GLN	+3.311	+3.184	
317	CYS135ASN	-3.122	-3.184	
318	ALA189ASN	+3.243	+3.181	
319	GLU285TYR	-3.134	-3.176	
320	GLY117MET	+3.105	+3.175	
321	GLY117VAL	+3.244	+3.172	
322	ALA161GLN	+3.296	+3.169	
323	GLU221PRO	-3.168	-3.168	
324	SER260PRO	-3.165	-3.165	
325	LYS120TRP	+3.101	+3.162	
326	ARG248MET	-3.231	-3.161	
327	ARG196GLY	-3.232	-3.157	
328	GLU171ALA	-3.305	-3.146	
329	LYS120PHE	+3.065	+3.141	
330	SER127GLY	-3.214	-3.139	
331	GLY279ILE	+3.327	+3.136	
332	CYS238GLN	-3.004	-3.130	
333	ARG249ALA	-3.285	-3.126	

#	Mutation	$\Delta\Delta G_p$	$\Delta\Delta G_c$	Refs
334	ARG283ALA	-3.278	-3.118	
335	CYS135GLY	-3.180	-3.105	
336	SER185TRP	-3.156	-3.095	
337	THR102PRO	-3.093	-3.093	
338	CYS135GLN	-2.958	-3.084	
339	ARG174LEU	-3.090	-3.084	
340	LYS139MET	+3.015	+3.084	
341	PHE134LYS	-3.053	-3.069	
342	GLY117HIS	+3.063	+3.066	
343	ASP259PRO	-3.059	-3.059	
344	ARG273VAL	+3.131	+3.058	
345	GLY262ARG	+3.181	+3.054	
346	ALA138PRO	-3.052	-3.052	
347	TYR126ALA	-3.207	-3.047	
348	GLU285GLN	-2.918	-3.045	
349	VAL225ASN	+3.105	+3.043	
350	VAL272SER	-3.046	-3.041	
351	LEU289TRP	+2.978	+3.039	
352	THR284GLY	-3.114	-3.039	
353	GLY105ASP	+3.014	+3.036	
354	ARG175PHE	-3.112	-3.036	
355	ARG175GLU	-3.057	-3.036	
356	SER185ALA	-3.194	-3.034	
357	SER106PRO	-3.024	-3.024	
358	SER240PHE	+2.944	+3.020	
359	GLY108ILE	+3.200	+3.010	
360	VAL272ALA	-3.163	-3.003	
361	ASP281LYS	-2.979	-2.995	
362	CYS242TYR	-2.952	-2.994	
363	VAL122GLY	-3.068	-2.993	
364	TYR103LYS	-2.977	-2.993	
365	HIS193MET	+2.915	+2.985	
366	ALA119PHE	+2.908	+2.984	
367	SER185HIS	-2.985	-2.982	
368	LEU133ALA	-3.142	-2.982	
369	GLU171PRO	-2.978	-2.978	
370	SER183ALA	-3.138	-2.978	
371	SER227PRO	-2.976	-2.976	
372	TYR236VAL	+3.045	+2.972	
373	ASN288TRP	+2.911	+2.972	
374	THR150PRO	-2.966	-2.966	
375	ILE255ALA	-3.121	-2.961	
376	SER127PHE	+2.881	+2.957	
377	PHE109PRO	-2.952	-2.952	
378	ALA138HIS	+2.941	+2.944	
379	ALA189LYS	+2.959	+2.943	
380	LEU114PRO	-2.936	-2.936	
381	ALA159PHE	+2.854	+2.929	
382	ILE255GLN	-2.795	-2.922	
383	TYR163TRP	-2.980	-2.919	
384	HIS168CYS	+2.847	+2.919	
385	THR125CYS	+2.844	+2.916	
386	CYS277GLY	-2.990	-2.915	

#	Mutation	$\Delta\Delta G_p$	$\Delta\Delta G_c$	Refs
387	ALA129MET	+2.840	+2.909	
388	ARG248ASN	-2.845	-2.907	
389	ALA119MET	+2.836	+2.906	
390	GLY117TRP	+2.839	+2.900	
391	ALA203HIS	+2.896	+2.899	
392	GLU271GLY	-2.972	-2.897	
393	ALA119ILE	+3.086	+2.895	
394	TYR126GLN	-2.768	-2.895	
395	GLN144GLY	-2.969	-2.894	
396	TYR103GLY	-2.967	-2.892	
397	ARG213LYS	-2.876	-2.892	
398	SER96ARG	+3.018	+2.891	
399	ALA119ARG	+3.015	+2.887	
400	TYR236LEU	+2.877	+2.884	
401	ARG174ASP	-2.899	-2.877	
402	LEU137TRP	+2.811	+2.872	
403	ARG283GLY	-2.943	-2.868	
404	ALA119CYS	+2.790	+2.862	
405	SER240PRO	-2.860	-2.860	
406	ALA159HIS	+2.854	+2.858	
407	ILE195GLY	-2.933	-2.858	
408	CYS124LEU	-2.863	-2.857	
409	THR140PRO	-2.851	-2.851	
410	LEU133GLU	+2.828	+2.849	
411	CYS238LYS	-2.833	-2.849	
412	ALA189LEU	+2.843	+2.849	
413	CYS242LYS	-2.831	-2.847	
414	HIS115PRO	-2.845	-2.845	
415	ALA161CYS	+2.769	+2.841	
416	GLY117ARG	+2.967	+2.840	
417	ILE232TYR	+2.882	+2.840	
418	HIS179ARG	+2.965	+2.837	
419	LYS139ILE	+3.028	+2.837	
420	ALA276PHE	+2.761	+2.837	
421	ARG283PRO	-2.834	-2.834	
422	LEU194GLY	-2.907	-2.832	
423	ILE255ASN	-2.768	-2.829	
424	ARG290TRP	+2.767	+2.828	
425	GLY112CYS	+2.750	+2.822	
426	SER185ILE	-2.630	-2.821	
427	HIS168LEU	+2.815	+2.821	
428	CYS277ALA	-2.975	-2.815	
429	TYR163MET	-2.884	-2.815	
430	LEU264ARG	+2.935	+2.808	
431	LYS132GLY	-2.881	-2.805	
432	SER127VAL	+2.876	+2.803	
433	VAL197GLY	-2.877	-2.802	
434	ARG158GLN	-2.673	-2.800	
435	SER127PRO	-2.789	-2.789	
436	GLU286ALA	-2.947	-2.788	
437	GLY266ILE	+2.970	+2.780	
438	GLN136CYS	+2.703	+2.774	
439	ALA203LEU	+2.768	+2.774	

#	Mutation	$\Delta\Delta G_p$	$\Delta\Delta G_c$	Refs
440	ASN200MET	+2.702	+2.772	
441	TYR163ASN	-2.707	-2.769	
442	ASP281GLN	-2.640	-2.767	
443	GLU204MET	+2.697	+2.766	
444	GLY266PRO	-2.766	-2.766	
445	ARG280GLY	-2.840	-2.765	
446	SER121TRP	+2.696	+2.757	
447	ARG181TYR	-2.711	-2.753	
448	LEU137MET	+2.678	+2.747	
449	GLY226ASN	+2.809	+2.747	
450	ARG174HIS	-2.747	-2.743	
451	PHE109ARG	+2.865	+2.738	
452	CYS238ASN	-2.674	-2.736	
453	CYS176ARG	-2.607	-2.735	
454	TYR126LYS	-2.716	-2.732	
455	VAL122THR	-2.752	-2.723	
456	CYS124TRP	-2.782	-2.721	
457	ALA161VAL	+2.793	+2.721	
458	ALA119HIS	+2.716	+2.719	
459	ILE255ARG	-2.591	-2.718	
460	CYS277LYS	-2.701	-2.718	
461	TYR163HIS	-2.721	-2.717	
462	GLN192ARG	+2.843	+2.716	
463	CYS275GLY	-2.790	-2.715	
464	GLN136LYS	-2.699	-2.715	
465	ILE255ASP	-2.736	-2.714	
466	VAL272THR	-2.743	-2.713	
467	ALA203ASP	+2.690	+2.712	
468	ARG248SER	-2.717	-2.712	
469	ARG280ALA	-2.867	-2.707	
470	ASN288MET	+2.630	+2.700	
471	VAL122LEU	-2.704	-2.699	
472	GLY108TYR	+2.740	+2.698	
473	SER185PHE	-2.772	-2.696	
474	ARG280PRO	-2.695	-2.695	
475	VAL143GLY	-2.769	-2.694	
476	CYS220ALA	-2.850	-2.690	
477	LYS139PHE	+2.613	+2.689	
478	LYS164ALA	-2.847	-2.687	
479	LEU252LYS	-2.670	-2.686	
480	LEU265ARG	+2.813	+2.686	
481	ARG248HIS	-2.688	-2.685	
482	THR211GLN	-2.554	-2.681	
483	ALA119VAL	+2.753	+2.680	
484	ARG248ILE	-2.486	-2.676	
485	VAL97ALA	-2.835	-2.675	
486	TYR236GLY	-2.746	-2.671	
487	TYR107PRO	-2.668	-2.668	
488	ASP281TYR	-2.624	-2.666	
489	GLU204LEU	+2.659	+2.665	
490	ALA129TRP	+2.604	+2.665	
491	CYS176TYR	-2.620	-2.662	
492	ARG175THR	-2.691	-2.661	

#	Mutation	$\Delta\Delta G_p$	$\Delta\Delta G_c$	Refs
493	ALA203GLU	+2.636	+2.658	
494	LEU264HIS	+2.651	+2.654	
495	ARG110MET	+2.579	+2.648	
496	GLY279GLU	+2.627	+2.648	
497	CYS124THR	-2.673	-2.643	
498	ASP268LYS	-2.625	-2.641	
499	ILE195ALA	-2.800	-2.640	
500	ARG181GLN	-2.511	-2.638	
501	ALA189MET	+2.566	+2.635	
502	ALA161ASP	+2.611	+2.634	
503	ARG213SER	-2.637	-2.632	
504	LEU133ARG	+2.758	+2.631	
505	HIS168ILE	+2.821	+2.631	
506	SER185VAL	-2.553	-2.625	
507	SER121PHE	+2.549	+2.625	
508	GLY112PHE	+2.548	+2.624	
509	ASP207ASN	+2.686	+2.624	
510	ASN131PRO	-2.620	-2.620	
511	ALA203MET	+2.547	+2.617	
512	THR230HIS	-2.619	-2.616	
513	THR231ALA	-2.775	-2.615	
514	ASN235GLU	+2.594	+2.615	
515	SER241PRO	-2.608	-2.608	
516	THR140CYS	+2.537	+2.608	
517	ILE254PRO	-2.605	-2.605	
518	LEU133HIS	+2.600	+2.603	
519	MET237CYS	+2.527	+2.599	
520	LEU289PHE	+2.522	+2.598	
521	SER183ASP	-2.619	-2.597	
522	CYS238PHE	-2.664	-2.588	
523	TYR234GLY	-2.662	-2.587	
524	ARG282CYS	+2.510	+2.582	
525	PHE212PRO	-2.579	-2.579	
526	LEU130HIS	+2.572	+2.575	
527	SER260LEU	-2.579	-2.573	
528	LYS120ARG	+2.700	+2.573	
529	SER185MET	-2.642	-2.573	
530	LYS164GLY	-2.644	-2.569	
531	CYS275TRP	-2.629	-2.568	
532	ILE162GLU	-2.585	-2.564	
533	TYR234SER	-2.568	-2.563	
534	ARG110PHE	+2.486	+2.562	
535	TYR163GLU	-2.579	-2.558	
536	LEU206GLN	-2.429	-2.556	
537	CYS275ILE	-2.362	-2.553	
538	ARG283LEU	-2.558	-2.552	
539	GLY108MET	+2.477	+2.546	
540	ILE232HIS	+2.541	+2.544	
541	CYS275LEU	-2.549	-2.543	
542	SER99PRO	-2.542	-2.542	
543	ARG158ASP	-2.559	-2.537	
544	CYS238ASP	-2.559	-2.537	
545	CYS238HIS	-2.540	-2.537	

#	Mutation	$\Delta\Delta G_p$	$\Delta\Delta G_c$	Refs
546	VAL172GLN	-2.410	-2.537	
547	ARG273PHE	+2.457	+2.533	
548	ARG209PRO	-2.530	-2.530	
549	GLY105ILE	+2.713	+2.522	
550	PHE134GLY	-2.595	-2.521	
551	CYS238TYR	-2.478	-2.520	
552	SER269PRO	-2.517	-2.517	
553	VAL157TYR	+2.554	+2.512	
554	VAL97MET	-2.579	-2.509	
555	GLY117GLU	+2.485	+2.506	
556	ALA189PRO	-2.503	-2.503	
557	HIS193TYR	+2.542	+2.500	
558	GLY245ARG	+2.621	+2.494	
559	LEU206ILE	-2.303	-2.494	
560	ALA189GLU	+2.472	+2.493	
561	THR118PRO	-2.493	-2.493	
562	GLY245PHE	+2.413	+2.489	
563	GLY108ARG	+2.615	+2.487	
564	MET160TYR	+2.528	+2.486	
565	LYS139PRO	-2.484	-2.484	
566	HIS193CYS	+2.412	+2.483	
567	TRP146TYR	+2.524	+2.482	
568	ASP228PRO	-2.479	-2.479	
569	ALA138THR	+2.448	+2.478	
570	LEU137PHE	+2.401	+2.477	
571	LYS120ILE	+2.667	+2.477	
572	GLY112GLU	+2.452	+2.473	
573	LEU252TRP	-2.525	-2.465	
574	ILE195ASP	-2.485	-2.462	
575	ASP207HIS	+2.459	+2.462	
576	ARG249ASN	-2.395	-2.457	
577	ARG213THR	-2.478	-2.448	
578	HIS168LYS	+2.461	+2.445	
579	ALA161ILE	+2.634	+2.443	
580	HIS168THR	+2.413	+2.442	
581	CYS277SER	-2.442	-2.438	
582	ILE255THR	-2.466	-2.437	
583	GLU271SER	-2.441	-2.437	
584	GLY266ASP	+2.414	+2.436	
585	ARG181CYS	-2.507	-2.436	
586	ARG248VAL	-2.363	-2.435	
587	GLY112TYR	+2.477	+2.435	
588	CYS277ASN	-2.370	-2.432	
589	ARG174ASN	-2.369	-2.430	
590	ARG110VAL	+2.501	+2.429	
591	GLN192ALA	-2.585	-2.425	
592	LEU264GLU	+2.402	+2.423	
593	ARG174VAL	-2.350	-2.422	
594	GLY112ILE	+2.608	+2.417	
595	SER185TYR	-2.374	-2.416	
596	CYS275VAL	-2.337	-2.409	
597	LYS139ARG	+2.534	+2.407	
598	GLY245TYR	+2.441	+2.399	

#	Mutation	$\Delta\Delta G_p$	$\Delta\Delta G_c$	Refs
599	ARG273LEU	+2.387	+2.393	
600	SER240LEU	+2.387	+2.393	

Appendix I: 1SPD (SOD1) whole-chain scan, top 400 (bias-corrected)

Full single-mutant scan of 1SPD chain A (153 positions, 2907 rows, WT=PRO excluded), ranked by $|\Delta\Delta G_{\text{corr}}|$. Source: build/scan_1spd_whole_v2.json. SOD1 is out-of-family for the model (Section on generalization gradient); these values are reported as rank-order hypotheses only.

#	Mutation	$\Delta\Delta G_{\text{pred}}$	$\Delta\Delta G_{\text{corr}}$	Refs
1	GLY82ASN	+5.878	+5.816	
2	GLY33TYR	+5.738	+5.696	
3	ILE112PRO	-5.556	-5.556	
4	CYS6PRO	-5.441	-5.441	
5	GLY33TRP	+5.277	+5.338	
6	PHE64PRO	-5.317	-5.317	
7	ASN65PRO	-5.315	-5.315	
8	PHE45PRO	-5.164	-5.164	
9	GLY82LYS	+5.157	+5.141	
10	HIS63PRO	-5.126	-5.126	
11	GLY41TRP	+5.001	+5.062	
12	GLY82HIS	+4.932	+4.935	
13	ILE112GLY	-4.947	-4.872	
14	ASP83PRO	-4.868	-4.868	
15	GLY41ILE	+5.044	+4.853	
16	GLY82LEU	+4.824	+4.830	
17	ALA4TYR	+4.866	+4.824	
18	LEU84PRO	-4.764	-4.764	
19	GLY82ARG	+4.756	+4.629	
20	HIS80PRO	-4.557	-4.557	
21	CYS6ASN	-4.483	-4.545	
22	VAL81PRO	-4.490	-4.490	
23	GLY41PHE	+4.385	+4.460	
24	ILE99PRO	-4.404	-4.404	
25	ALA4TRP	+4.314	+4.375	
26	ILE104PRO	-4.344	-4.344	
27	CYS6ALA	-4.482	-4.322	
28	ILE149PRO	-4.311	-4.311	
29	GLY147CYS	+4.223	+4.295	
30	ASN86PRO	-4.291	-4.291	
31	VAL5PRO	-4.267	-4.267	
32	HIS46PRO	-4.255	-4.255	
33	ASN19TYR	+4.214	+4.172	
34	CYS6SER	-4.171	-4.166	
35	ARG115PRO	-4.150	-4.150	
36	ALA145TYR	+4.186	+4.144	

#	Mutation	$\Delta\Delta G_{\text{pred}}$	$\Delta\Delta G_{\text{corr}}$	Refs
37	GLY82THR	+4.109	+4.139	
38	ALA145CYS	+4.058	+4.130	
39	HIS80ALA	-4.273	-4.113	
40	ALA4LEU	+4.098	+4.103	
41	ALA4CYS	+3.976	+4.048	
42	GLY127LEU	+4.007	+4.013	
43	GLY82ILE	+4.176	+3.986	
44	ALA60ARG	+4.112	+3.985	
45	GLY33PHE	+3.905	+3.981	
46	HIS80GLY	-4.055	-3.980	
47	GLY16CYS	+3.904	+3.976	
48	ARG115HIS	-3.954	-3.950	
49	THR137TRP	+3.868	+3.929	
50	VAL47PRO	-3.878	-3.878	
51	SER105PRO	-3.870	-3.870	
52	CYS6HIS	-3.871	-3.868	
53	GLY51ARG	+3.994	+3.866	
54	LEU8PRO	-3.862	-3.862	
55	GLY44ILE	+4.043	+3.852	
56	LEU117GLY	-3.910	-3.835	
57	GLY82TYR	+3.862	+3.820	
58	PHE20GLY	-3.889	-3.814	
59	VAL47ARG	+3.939	+3.812	
60	LYS30PRO	-3.803	-3.803	
61	ILE113PRO	-3.794	-3.794	
62	GLY82SER	+3.714	+3.718	
63	GLY82GLU	+3.688	+3.709	
64	HIS120PRO	-3.692	-3.692	
65	ALA145LEU	+3.672	+3.678	
66	GLY51ILE	+3.867	+3.676	
67	GLY127HIS	+3.668	+3.671	
68	TRP32GLY	-3.733	-3.658	
69	GLY37ILE	+3.839	+3.648	
70	ALA152TYR	+3.683	+3.641	
71	CYS6GLY	-3.708	-3.633	
72	GLY85PRO	-3.625	-3.625	
73	GLY41TYR	+3.666	+3.624	
74	GLY33CYS	+3.544	+3.616	
75	GLY82PHE	+3.531	+3.607	
76	SER34TYR	+3.635	+3.593	
77	HIS80ASP	-3.598	-3.576	
78	ALA4MET	+3.476	+3.545	
79	THR137ILE	+3.725	+3.534	
80	GLY12CYS	+3.426	+3.498	
81	ALA4PHE	+3.422	+3.498	
82	THR135HIS	+3.487	+3.490	
83	GLY37LEU	+3.478	+3.484	

#	Mutation	$\Delta\Delta G_{\text{pred}}$	$\Delta\Delta G_{\text{corr}}$	Refs
84	VAL119ILE	+3.669	+3.478	
85	GLY16TYR	+3.516	+3.474	
86	CYS111PRO	-3.469	-3.469	
87	GLY93LEU	+3.459	+3.465	
88	HIS80SER	-3.451	-3.446	
89	HIS80MET	-3.492	-3.423	
90	VAL87PRO	-3.414	-3.414	
91	GLY127TRP	+3.344	+3.405	
92	ILE18TYR	+3.444	+3.402	
93	LEU126SER	-3.406	-3.401	
94	GLY127ILE	+3.564	+3.373	
95	GLU49ARG	+3.500	+3.373	
96	ALA123ARG	+3.499	+3.372	
97	GLY147ILE	+3.557	+3.366	
98	GLY114CYS	+3.290	+3.362	
99	ARG143PRO	-3.361	-3.361	
100	GLY147TYR	+3.403	+3.361	
101	LEU126THR	-3.389	-3.359	
102	THR88GLY	-3.433	-3.357	
103	ASN19TRP	+3.297	+3.357	
104	TRP32ALA	-3.514	-3.354	
105	VAL47SER	+3.345	+3.349	
106	ILE149HIS	-3.335	-3.332	
107	SER102PRO	-3.329	-3.329	
108	VAL31PRO	-3.324	-3.324	
109	ILE99GLY	-3.390	-3.315	
110	CYS111TYR	+3.356	+3.314	
111	GLY150CYS	+3.241	+3.312	
112	GLY147LEU	+3.300	+3.306	
113	VAL119LEU	+3.296	+3.302	
114	THR116GLY	-3.368	-3.293	
115	HIS110PRO	-3.277	-3.277	
116	LEU8LYS	-3.253	-3.270	
117	GLY41VAL	+3.339	+3.267	
118	GLY93TRP	+3.203	+3.263	
119	VAL29PRO	-3.248	-3.248	
120	CYS6GLN	-3.112	-3.239	
121	TRP32SER	-3.242	-3.237	
122	GLY51CYS	+3.156	+3.228	
123	ASN53ARG	+3.349	+3.222	
124	ILE112LYS	-3.204	-3.220	
125	LEU144TRP	+3.154	+3.215	
126	CYS6LYS	-3.198	-3.215	
127	ILE149ALA	-3.369	-3.209	
128	VAL94GLY	-3.271	-3.196	
129	GLY33THR	+3.156	+3.186	
130	THR135TRP	+3.124	+3.185	

#	Mutation	$\Delta\Delta G_{\text{pred}}$	$\Delta\Delta G_{\text{corr}}$	Refs
131	ALA152TRP	+3.118	+3.179	
132	CYS146PRO	-3.170	-3.170	
133	ILE112GLN	-3.039	-3.166	
134	ARG115GLY	-3.240	-3.165	
135	HIS71SER	-3.162	-3.157	
136	ILE112ASP	-3.176	-3.154	
137	PHE20PRO	-3.152	-3.152	
138	ALA123PHE	+3.074	+3.150	
139	GLY12ILE	+3.337	+3.146	
140	VAL118ILE	+3.336	+3.145	
141	ILE149ASN	-3.078	-3.139	
142	ASN19PHE	+3.062	+3.138	
143	ALA60CYS	+3.065	+3.137	
144	HIS71GLY	-3.208	-3.133	
145	GLY114TYR	+3.159	+3.118	
146	GLY82PRO	-3.115	-3.115	
147	ILE112GLU	-3.128	-3.107	
148	HIS71THR	-3.134	-3.105	
149	GLN22PRO	-3.087	-3.087	
150	LEU126VAL	-3.013	-3.086	
151	SER59ARG	+3.202	+3.075	
152	ASP11TYR	+3.097	+3.055	
153	GLY10CYS	+2.983	+3.054	
154	GLY12TYR	+3.096	+3.054	
155	VAL5TRP	+2.992	+3.053	
156	GLY37TRP	+2.988	+3.049	
157	GLY73TRP	+2.986	+3.047	
158	ALA145TRP	+2.986	+3.047	
159	HIS63GLY	-3.117	-3.042	
160	ALA152LEU	+3.028	+3.034	
161	LEU126ASP	-3.046	-3.024	
162	GLY44PHE	+2.947	+3.023	
163	SER98TYR	+3.062	+3.020	
164	GLY72LEU	+3.011	+3.017	
165	VAL5TYR	+3.037	+2.995	
166	GLU100PRO	-2.994	-2.994	
167	THR137HIS	+2.988	+2.991	
168	ALA4LYS	+3.004	+2.987	
169	GLY85ARG	+3.113	+2.986	
170	HIS43PRO	-2.977	-2.977	
171	GLY114TRP	+2.903	+2.964	
172	LEU67TYR	+3.002	+2.960	
173	ALA60THR	+2.927	+2.956	
174	ALA4ARG	+3.082	+2.954	
175	ASP90TRP	+2.893	+2.953	
176	GLY12LEU	+2.947	+2.953	
177	TRP32LYS	-2.932	-2.948	

#	Mutation	$\Delta\Delta G_{\text{pred}}$	$\Delta\Delta G_{\text{corr}}$	Refs
178	SER134ILE	+3.128	+2.937	
179	HIS71MET	-3.001	-2.932	
180	THR58ILE	+3.105	+2.914	
181	LEU144TYR	+2.955	+2.913	
182	VAL87GLY	-2.986	-2.911	
183	LEU126LYS	-2.889	-2.906	
184	CYS57MET	-2.975	-2.905	
185	GLU77CYS	+2.828	+2.900	
186	CYS6GLU	-2.920	-2.899	
187	PHE45GLY	-2.969	-2.893	
188	ALA60TYR	+2.932	+2.890	
189	HIS63GLN	-2.763	-2.890	
190	LEU8HIS	-2.890	-2.887	
191	ALA145ARG	+3.011	+2.884	
192	THR88TRP	+2.819	+2.880	
193	ILE112ALA	-3.039	-2.879	
194	ILE99CYS	-2.947	-2.875	
195	ASP52ARG	+3.001	+2.874	
196	GLY37VAL	+2.946	+2.874	
197	GLY72TRP	+2.812	+2.873	
198	VAL14ASP	-2.893	-2.871	
199	GLY73CYS	+2.794	+2.866	
200	VAL14PRO	-2.863	-2.863	
201	VAL119TYR	+2.902	+2.860	
202	ALA4VAL	+2.931	+2.858	
203	ARG115TRP	-2.906	-2.845	
204	GLY51THR	+2.813	+2.842	
205	LEU126MET	-2.912	-2.842	
206	GLY93ILE	+3.032	+2.841	
207	LEU117PRO	-2.833	-2.833	
208	THR135ILE	+3.018	+2.827	
209	GLY16LEU	+2.809	+2.815	
210	SER107PRO	-2.796	-2.796	
211	GLY138TRP	+2.733	+2.793	
212	ASN65MET	-2.854	-2.784	
213	ILE99ASP	-2.803	-2.781	
214	ALA152ARG	+2.895	+2.768	
215	LEU8GLU	-2.784	-2.763	
216	HIS48PRO	-2.763	-2.763	
217	LEU84ALA	-2.917	-2.757	
218	HIS43GLY	-2.830	-2.755	
219	GLY147ARG	+2.880	+2.752	
220	GLY44PRO	-2.746	-2.746	
221	CYS6THR	-2.770	-2.741	
222	LEU144ILE	+2.930	+2.740	
223	GLY33HIS	+2.736	+2.739	
224	GLY41MET	+2.666	+2.736	

#	Mutation	$\Delta\Delta G_{\text{pred}}$	$\Delta\Delta G_{\text{corr}}$	Refs
225	GLY127GLU	+2.712	+2.733	
226	GLY33ARG	+2.856	+2.729	
227	GLY150TRP	+2.659	+2.720	
228	GLY82GLN	+2.841	+2.714	
229	ALA4ILE	+2.903	+2.713	
230	GLY93TYR	+2.755	+2.713	
231	VAL5CYS	+2.641	+2.712	
232	GLY73LEU	+2.705	+2.711	
233	SER68HIS	+2.698	+2.701	
234	GLY127PRO	+2.690	+2.690	
235	THR39ILE	+2.878	+2.687	
236	GLY150LEU	+2.678	+2.684	
237	THR88TYR	+2.725	+2.683	
238	GLY138LEU	+2.677	+2.683	
239	GLY114LEU	+2.677	+2.683	
240	CYS57GLN	-2.554	-2.680	
241	GLY72HIS	+2.673	+2.676	
242	GLY114MET	+2.598	+2.667	
243	GLU121ARG	+2.792	+2.665	
244	GLY114ILE	+2.853	+2.662	
245	HIS63ALA	-2.820	-2.660	
246	TRP32ILE	-2.460	-2.651	
247	ILE112ASN	-2.587	-2.649	
248	ALA145ILE	+2.839	+2.648	
249	PHE50ARG	+2.774	+2.647	
250	VAL81HIS	+2.644	+2.647	
251	GLY85TYR	+2.688	+2.646	
252	ILE18PRO	-2.641	-2.641	
253	ASP101PRO	-2.641	-2.641	
254	LYS128ILE	+2.830	+2.639	
255	VAL119ARG	+2.766	+2.639	
256	PHE20SER	-2.640	-2.635	
257	GLY44ARG	+2.758	+2.630	
258	LEU8ASN	-2.567	-2.628	
259	LYS128HIS	+2.624	+2.628	
260	LEU126GLU	-2.643	-2.621	
261	VAL118ARG	+2.748	+2.621	
262	CYS146GLY	-2.695	-2.620	
263	VAL47THR	+2.588	+2.618	
264	THR116PRO	-2.616	-2.616	
265	VAL118THR	+2.582	+2.611	
266	GLY51TYR	+2.650	+2.608	
267	ASP83GLY	-2.680	-2.605	
268	HIS63MET	-2.671	-2.602	
269	ALA60ILE	+2.792	+2.601	
270	ILE112HIS	-2.604	-2.600	
271	ALA89TYR	+2.638	+2.596	

#	Mutation	$\Delta\Delta G_{\text{pred}}$	$\Delta\Delta G_{\text{corr}}$	Refs
272	ILE113HIS	-2.598	-2.595	
273	VAL119TRP	+2.534	+2.595	
274	GLY16PHE	+2.512	+2.588	
275	GLY41LEU	+2.582	+2.588	
276	SER142ARG	+2.708	+2.581	
277	GLY127GLN	+2.707	+2.580	
278	LYS9TYR	+2.621	+2.579	
279	LEU126ARG	-2.444	-2.571	
280	VAL94ASP	-2.592	-2.570	
281	ASP11TRP	+2.509	+2.570	
282	ASN65CYS	-2.641	-2.569	
283	ASP11CYS	+2.497	+2.569	
284	GLY150TYR	+2.607	+2.565	
285	LEU67SER	+2.559	+2.564	
286	GLY127ASN	+2.624	+2.563	
287	LEU8GLY	-2.636	-2.561	
288	VAL148CYS	+2.485	+2.557	
289	ASN86GLY	-2.629	-2.554	
290	GLY51SER	+2.549	+2.554	
291	ASP83MET	-2.621	-2.552	
292	GLY33GLU	+2.521	+2.542	
293	LYS122PHE	+2.465	+2.541	
294	GLY73ARG	+2.664	+2.537	
295	ALA123HIS	+2.534	+2.537	
296	GLU121PRO	-2.535	-2.535	
297	GLY138ILE	+2.725	+2.534	
298	ASP124PRO	-2.528	-2.528	
299	CYS6ASP	-2.550	-2.528	
300	ILE149GLY	-2.601	-2.526	
301	ALA123ILE	+2.715	+2.524	
302	GLY130TRP	+2.455	+2.516	
303	ILE99GLN	-2.382	-2.509	
304	LEU67PHE	+2.433	+2.509	
305	CYS146GLN	-2.381	-2.508	
306	ARG143GLY	-2.580	-2.505	
307	ASP109PRO	-2.500	-2.500	
308	HIS71ASP	-2.521	-2.499	
309	GLY147TRP	+2.436	+2.497	
310	PHE20ALA	-2.656	-2.496	
311	GLU121PHE	+2.418	+2.494	
312	ILE35GLY	-2.564	-2.489	
313	TRP32VAL	-2.411	-2.484	
314	CYS57GLU	-2.505	-2.484	
315	ALA152PHE	+2.406	+2.482	
316	VAL87TRP	+2.418	+2.478	
317	LEU8GLN	-2.350	-2.476	
318	GLU49THR	+2.446	+2.476	

#	Mutation	$\Delta\Delta G_{\text{pred}}$	$\Delta\Delta G_{\text{corr}}$	Refs
319	LEU8ASP	-2.487	-2.465	
320	LEU8ALA	-2.622	-2.462	
321	ILE99HIS	-2.459	-2.456	
322	ILE18ALA	-2.615	-2.455	
323	CYS6VAL	-2.380	-2.453	
324	VAL103PRO	-2.443	-2.443	
325	ASN86TYR	+2.483	+2.441	
326	GLU49PRO	-2.441	-2.441	
327	SER98TRP	+2.379	+2.439	
328	TRP32PRO	-2.433	-2.433	
329	ASP96TYR	+2.469	+2.427	
330	CYS57ASP	-2.449	-2.426	
331	PHE45CYS	-2.498	-2.426	
332	ARG115ALA	-2.586	-2.426	
333	HIS46ARG	+2.553	+2.425	
334	LEU106PRO	-2.425	-2.425	
335	ILE151TRP	+2.357	+2.418	
336	SER34PHE	+2.340	+2.416	
337	LEU67PRO	+2.416	+2.416	
338	CYS57PRO	-2.413	-2.413	
339	VAL148ARG	+2.540	+2.413	
340	GLY12VAL	+2.485	+2.412	
341	THR135LEU	+2.405	+2.410	
342	GLY41ALA	+2.244	+2.404	
343	PHE45MET	-2.472	-2.402	
344	VAL148PRO	-2.401	-2.401	
345	TRP32GLU	-2.417	-2.396	
346	VAL47LEU	+2.389	+2.395	
347	VAL148TYR	+2.437	+2.395	
348	TRP32MET	-2.463	-2.394	
349	ILE113LYS	-2.378	-2.394	
350	HIS120GLY	-2.469	-2.393	
351	THR137LEU	+2.387	+2.393	
352	VAL118TYR	+2.433	+2.391	
353	LEU126PHE	-2.466	-2.390	
354	THR58TRP	+2.329	+2.389	
355	ASP124VAL	-2.313	-2.385	
356	HIS110GLY	-2.458	-2.382	
357	VAL97PRO	-2.381	-2.381	
358	GLY51TRP	+2.317	+2.378	
359	GLY56MET	-2.445	-2.375	
360	LYS136TRP	+2.314	+2.375	
361	GLY33ASN	+2.433	+2.372	
362	VAL5SER	-2.375	-2.371	
363	GLY44TYR	+2.412	+2.370	
364	GLU49SER	+2.363	+2.368	
365	ILE99GLU	-2.386	-2.365	

#	Mutation	$\Delta\Delta G_{\text{pred}}$	$\Delta\Delta G_{\text{corr}}$	Refs
366	LYS70ILE	+2.550	+2.359	
367	ASP83ALA	-2.516	-2.357	
368	ASN19CYS	+2.282	+2.354	
369	ARG143LYS	-2.337	-2.353	
370	LEU38TRP	+2.290	+2.351	
371	ILE99SER	-2.355	-2.350	
372	VAL94MET	-2.419	-2.350	
373	VAL5GLY	-2.424	-2.349	
374	ILE112MET	-2.413	-2.343	
375	VAL94LYS	-2.325	-2.341	
376	GLY93ASN	+2.397	+2.335	
377	ASP83ARG	+2.462	+2.335	
378	GLY44LEU	+2.328	+2.334	
379	VAL119PRO	-2.330	-2.330	
380	ILE113ALA	-2.487	-2.327	
381	LEU67HIS	+2.323	+2.326	
382	ILE99ALA	-2.486	-2.326	
383	VAL14HIS	-2.324	-2.321	
384	HIS80GLN	-2.194	-2.321	
385	VAL7PRO	-2.311	-2.311	
386	PHE64MET	-2.381	-2.311	
387	GLY114ARG	+2.438	+2.311	
388	CYS57LYS	-2.293	-2.309	
389	GLY141PHE	+2.233	+2.309	
390	HIS46THR	+2.276	+2.306	
391	LYS128TRP	+2.245	+2.306	
392	ASP52PRO	-2.300	-2.300	
393	LEU8SER	-2.304	-2.299	
394	TRP32LEU	-2.299	-2.293	
395	GLY93GLN	+2.418	+2.291	
396	ASP90TYR	+2.330	+2.288	
397	GLY72ASN	+2.350	+2.288	
398	ARG143GLN	-2.159	-2.286	
399	GLY72ARG	+2.411	+2.284	
400	ILE35HIS	-2.287	-2.284	

Appendix J: 2VUK per-position aggregation

Each of the 195 scanned positions of 2VUK chain A, aggregated over its 19 mutants (WT=PRO rows excluded): mean corrected $\Delta\Delta G$, mean $|\Delta\Delta G_{\text{corr}}|$, and max $|\Delta\Delta G_{\text{corr}}|$. Sorted by the last column - position-level destabilisation hotspots. Source: build/scan_2vuk_whole_v2.json.

#	Pos	WT	mean $\Delta\Delta G_c$	mean $ \Delta\Delta G_c $	max $ \Delta\Delta G_c $
1	158	ARG	-1.313	1.686	9.473
2	162	ILE	-4.072	4.078	8.668
3	255	ILE	-2.267	2.337	8.438

#	Pos	WT	mean $\Delta\Delta G_c$	mean $ \Delta\Delta G_c $	max $ \Delta\Delta G_c $
4	126	TYR	-1.516	1.897	8.220
5	205	TYR	-0.795	1.212	8.143
6	175	ARG	-3.943	3.996	7.803
7	196	ARG	-1.176	1.653	7.643
8	197	VAL	-0.469	1.401	7.376
9	217	VAL	-0.648	1.221	7.355
10	272	VAL	-1.817	2.033	7.297
11	215	SER	+0.292	1.464	7.179
12	204	GLU	-0.301	1.291	7.107
13	157	VAL	+0.066	1.503	7.038
14	124	CYS	-2.933	2.933	6.997
15	176	CYS	-4.502	4.502	6.946
16	238	CYS	-3.463	3.463	6.944
17	174	ARG	-2.909	2.909	6.939
18	216	VAL	-0.916	1.414	6.905
19	234	TYR	-1.493	1.673	6.875
20	236	TYR	-0.101	1.778	6.795
21	195	ILE	-0.934	1.620	6.596
22	256	THR	-0.699	1.307	6.532
23	240	SER	+1.169	1.610	6.524
24	242	CYS	-4.158	4.158	6.514
25	163	TYR	-2.243	2.263	6.428
26	206	LEU	-1.035	1.430	6.294
27	160	MET	-0.201	1.260	6.186
28	144	GLN	-0.375	1.248	6.180
29	245	GLY	+1.194	1.636	6.052
30	246	MET	+1.273	1.551	6.019
31	202	ARG	-0.616	0.970	5.958
32	181	ARG	-3.968	3.968	5.946
33	273	ARG	+0.532	1.938	5.913
34	214	HIS	-0.286	1.308	5.884
35	194	LEU	-1.115	1.484	5.878
36	271	GLU	-1.222	1.444	5.822
37	213	ARG	-2.214	2.270	5.804
38	198	GLU	-1.363	1.394	5.716
39	275	CYS	-2.603	2.603	5.711
40	201	LEU	-1.523	1.595	5.628
41	252	LEU	-2.056	2.137	5.620
42	241	SER	-0.021	1.238	5.599
43	200	ASN	+0.230	1.191	5.561
44	173	VAL	-0.544	1.607	5.538
45	189	ALA	+2.225	2.488	5.481
46	251	ILE	-1.557	1.808	5.480
47	172	VAL	-1.063	1.722	5.475
48	218	VAL	-0.430	1.098	5.466
49	282	ARG	-1.098	1.803	5.433
50	145	LEU	-0.584	1.336	5.418

#	Pos	WT	mean $\Delta\Delta G_c$	mean $ \Delta\Delta G_c $	max $ \Delta\Delta G_c $
51	132	LYS	+0.221	1.485	5.412
52	113	PHE	-0.778	1.188	5.319
53	134	PHE	-1.632	1.826	5.269
54	135	CYS	-2.107	2.141	5.252
55	243	MET	-0.313	1.130	5.224
56	161	ALA	+2.723	2.895	5.169
57	247	ASN	+0.734	1.266	5.110
58	138	ALA	+2.282	2.643	5.083
59	141	CYS	-0.872	1.167	5.031
60	156	ARG	-1.077	1.177	4.988
61	193	HIS	+0.827	1.804	4.968
62	239	TYR	-0.241	1.327	4.907
63	122	VAL	-2.139	2.264	4.880
64	123	THR	+0.255	1.589	4.878
65	253	THR	+0.608	1.306	4.819
66	159	ALA	+1.631	2.138	4.816
67	281	ASP	-1.893	2.016	4.792
68	111	LEU	-0.620	1.007	4.763
69	203	ALA	+1.541	2.038	4.727
70	258	GLU	-0.698	0.985	4.697
71	285	GLU	-1.862	1.992	4.691
72	270	PHE	-0.616	1.086	4.610
73	164	LYS	-0.985	1.297	4.596
74	257	LEU	-0.419	1.119	4.547
75	231	THR	-0.267	1.163	4.538
76	143	VAL	+0.049	1.180	4.529
77	136	GLN	-0.311	1.573	4.522
78	112	GLY	+1.038	1.561	4.497
79	110	ARG	+0.708	1.487	4.318
80	244	GLY	+0.266	1.026	4.316
81	230	THR	-0.818	1.119	4.300
82	248	ARG	-2.172	2.348	4.300
83	125	THR	+0.124	1.558	4.291
84	279	GLY	+1.379	1.798	4.223
85	155	THR	-0.826	1.203	4.182
86	133	LEU	+0.369	1.502	4.178
87	129	ALA	+1.237	1.404	4.175
88	276	ALA	+0.971	1.480	4.175
89	267	ARG	-0.094	1.045	4.174
90	178	HIS	+0.439	0.908	4.152
91	139	LYS	+1.861	2.123	4.114
92	179	HIS	+0.478	1.281	4.098
93	137	LEU	+1.884	1.979	4.071
94	235	ASN	-0.217	0.988	4.047
95	114	LEU	+0.884	1.470	4.009
96	192	GLN	+0.200	1.393	3.965
97	220	CYS	-0.304	1.428	3.961

#	Pos	WT	mean $\Delta\Delta G_c$	mean $ \Delta\Delta G_c $	max $ \Delta\Delta G_c $
98	188	LEU	-0.529	0.768	3.960
99	237	MET	-0.129	1.235	3.949
100	211	THR	-0.872	1.390	3.906
101	207	ASP	+0.330	1.203	3.902
102	249	ARG	-1.242	1.378	3.836
103	180	GLU	+0.289	1.192	3.835
104	183	SER	-1.607	1.667	3.830
105	119	ALA	+1.853	1.853	3.770
106	277	CYS	-1.450	1.484	3.720
107	127	SER	+0.901	1.783	3.664
108	268	ASP	-0.805	1.185	3.626
109	108	GLY	+1.345	1.584	3.618
110	168	HIS	+1.751	1.866	3.600
111	171	GLU	-0.947	1.284	3.588
112	232	ILE	+0.967	1.584	3.570
113	186	ASP	-1.029	1.057	3.515
114	286	GLU	-0.975	1.340	3.502
115	233	HIS	+0.295	0.870	3.501
116	199	GLY	+0.478	0.931	3.488
117	289	LEU	+1.241	1.544	3.472
118	104	GLN	+0.411	1.191	3.447
119	208	ASP	-0.682	1.019	3.438
120	103	TYR	-1.136	1.347	3.419
121	274	VAL	-0.101	1.207	3.390
122	117	GLY	+1.675	1.816	3.380
123	212	PHE	-0.313	1.059	3.363
124	229	CYS	-0.395	0.932	3.353
125	149	SER	-0.237	0.961	3.268
126	185	SER	-2.137	2.137	3.251
127	140	THR	+0.846	1.250	3.234
128	146	TRP	+0.234	1.095	3.218
129	100	GLN	+0.010	0.954	3.215
130	147	VAL	-0.329	0.848	3.210
131	225	VAL	+0.799	1.064	3.195
132	221	GLU	+0.309	0.938	3.168
133	260	SER	-0.930	1.096	3.165
134	120	LYS	+1.489	1.597	3.162
135	283	ARG	-0.768	1.177	3.118
136	102	THR	+0.177	0.768	3.093
137	259	ASP	-0.681	1.021	3.059
138	262	GLY	+1.023	1.195	3.054
139	284	THR	-0.096	1.063	3.039
140	105	GLY	+0.686	1.159	3.036
141	106	SER	+0.361	0.775	3.024
142	227	SER	-0.930	1.017	2.976
143	288	ASN	+0.584	1.047	2.972
144	150	THR	-0.246	0.776	2.966

#	Pos	WT	mean $\Delta\Delta G_c$	mean $ \Delta\Delta G_c $	max $ \Delta\Delta G_c $
145	109	PHE	+0.308	1.200	2.952
146	96	SER	+0.246	1.059	2.891
147	115	HIS	-0.737	0.934	2.845
148	290	ARG	+0.699	0.905	2.828
149	264	LEU	+1.195	1.451	2.808
150	266	GLY	+0.605	1.289	2.780
151	280	ARG	-0.961	1.223	2.765
152	121	SER	+0.720	0.954	2.757
153	226	GLY	+1.058	1.212	2.747
154	265	LEU	+0.523	1.075	2.686
155	97	VAL	-0.469	0.761	2.675
156	107	TYR	-0.335	0.720	2.668
157	131	ASN	+0.244	0.867	2.620
158	254	ILE	-0.459	1.066	2.605
159	130	LEU	+0.571	1.071	2.575
160	99	SER	-0.985	1.286	2.542
161	209	ARG	-0.247	0.967	2.530
162	269	SER	+0.695	0.996	2.517
163	118	THR	+0.272	1.298	2.493
164	228	ASP	-0.642	0.795	2.479
165	169	MET	+0.246	1.186	2.381
166	263	ASN	-0.944	1.084	2.374
167	182	CYS	-1.070	1.070	2.257
168	170	THR	+0.156	0.942	2.238
169	154	GLY	+0.338	1.091	2.174
170	287	GLU	-0.313	1.189	2.144
171	165	GLN	-0.279	0.763	2.139
172	184	ASP	+0.258	0.937	2.130
173	101	LYS	+0.163	0.572	2.119
174	261	SER	-0.834	0.874	2.046
175	210	ASN	-0.068	0.764	2.030
176	166	SER	-0.142	1.025	2.026
177	116	SER	+0.821	1.112	1.978
178	148	ASP	+0.437	0.797	1.879
179	167	GLN	-0.282	0.631	1.714
180	224	GLU	+0.087	0.643	1.643
181	187	GLY	+0.075	0.726	1.543

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